

AI-BASED MENTAL HEALTH RISK PREDICTION AND GUIDANCE SYSTEM



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EXORDIUM



Recite with the name of your ALLAH Who created,
He made man from the clot of blood.
Recite, for your ALLAH is the Most Generous,
Who taught writing by the pen. Taught man what not.
Yes, undoubtedly, man transgresses
Because, he thought himself self-sufficient.
Undoubtedly, unto your ALLAH is the return.
Taught man what HE knew
Well, you see HIM WHO forbids.

Reference: Surah Al Alaq (Verse 1-9) **Al Quran**

Statement of Submission

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Abstract

This project introduces NEXORA, a digital health platform designed to make mental health evaluation, support, and guidance accessible to everyone. In modern society, psychological well-being is frequently overlooked due to societal stigma, busy routines, and limited access to specialized care. NEXORA addresses these barriers by combining a preliminary screening tool, an artificial intelligence chatbot, and a direct referral directory for mental health specialists in Lahore. Users complete the standardized PHQ-9 (Patient Health Questionnaire), which evaluates self-reported symptoms to estimate depression risk tiers ranging from minimal to severe. Based on this risk assessment, the platform offers tailored recommendations, immediate crisis support information for high-risk cases, and an interactive AI assistant powered by Botpress for ongoing conversational support. Furthermore, NEXORA streamlines the process of finding and booking appointments with licensed local psychologists. Built using Python, Flask, MySQL, and modern web standards, NEXORA bridges the gap between self-screening and professional psychological assistance through a simple, secure, and user-centered web application.

Introduction:

Mental health is just as vital to human well-being as physical health, yet millions of people navigate emotional distress, anxiety, and depression without timely intervention. NEXORA was created to make early mental health evaluation simple, private, and approachable. Rather than waiting until psychological distress becomes overwhelming, individuals can use NEXORA to check in on their emotional well-being from the comfort and privacy of their own personal devices.

The platform centers around a well-established self-assessment tool known as the PHQ-9, which translates subjective feelings into an easy-to-understand risk level. By giving users clear insights into their current emotional state, NEXORA helps remove the confusion and uncertainty surrounding mental health. The ultimate goal of this project is to empower individuals with early awareness, immediate guidance, and direct pathways to professional care ensuring everyone has a fair chance at receiving the support they need.

Motivation:

In today's fast-paced world, personal well-being often takes a backseat to work, studies, and daily obligations. Recognizing when one needs help can be difficult, and seeking professional therapy often presents high barriers such as difficulty finding qualified local specialists, fear of social judgment, or simply not knowing where to start.

NEXORA was motivated by the need for a non-judgmental, immediate, and accessible entry point into mental health awareness:

- **Instant Self-Awareness:** Unlike traditional clinical visits that require long waiting times, NEXORA provides instant evaluation results. Users can take an assessment in minutes and receive clear, actionable feedback right away.
- **Continuous Support:** Through an integrated conversational AI chatbot, users can ask questions, process their feelings, and receive empathetic guidance 24/7 without feeling pressured.
- **A Bridge to Human Care:** Technology should complement human expertise, not replace it. NEXORA uses smart matching algorithms to connect users especially those identified as high-risk directly with qualified psychologists in Lahore, making the transition from self-assessment to professional counseling as smooth as possible.

Problem Statement:

Despite growing awareness around mental health, significant gaps remain between experiencing emotional difficulties and receiving proper professional help:

- **Delayed Intervention:** Most individuals do not consult a professional until psychological distress becomes severe, largely because early symptoms go unnoticed or unmeasured.
- **Lack of Accessible Screening:** Standardized screening tools are often confined to clinical environments, preventing everyday individuals from assessing their mental health proactively.
- **Fragmented Referral Pathways:** Finding a licensed psychologist, checking their availability, and booking a consultation is frequently confusing and overwhelming for someone already experiencing emotional strain.

NEXORA addresses these exact challenges by offering a unified, privacy-focused platform that combines standardized screening, AI-guided support, and direct specialist booking in one continuous workflow.

Approach:

NEXORA brings together modern web technologies, smart assessment logic, and conversational AI to deliver a clear, user-friendly experience:

- **Interactive PHQ-9 Assessment:** Users complete a simple 9-question survey. The system calculates a total score (0 to 27) and categorizes the result into clear risk tiers—Minimal, Mild, Moderate, Moderately Severe, or Severe.
- **Smart Risk Response & Crisis Alerts:** If a user's assessment indicates high risk or severe distress, the platform immediately prioritizes their safety by displaying emergency helpline information and prompting immediate connections with specialists.
- **AI-Powered Chatbot Integration:** Embedded directly within the interface, a custom Botpress chatbot offers continuous conversational support. It answers common health queries, explains assessment outcomes, and provides empathetic guidance tailored to the user's situation.
- **Psychologist Directory & Appointment Booking:** Users can browse a verified directory of mental health professionals in Lahore, view their credentials and availability, and schedule appointments directly through the platform.
- **Secure & Standardized Architecture:** Built on a Flask backend and MySQL database, NEXORA handles user data with care, employing secure password hashing, structured API routes, and clean database management to ensure complete privacy and reliability.

Chapter No. 1
INTRODUCTION

1.1 - Background and Motivation:

Mental health is a fundamental aspect of overall human well-being; however, it remains one of the most overlooked areas of public health, especially in developing countries. According to the World Health Organization, depression is among the leading causes of disability worldwide, affecting approximately 280 million people globally. In Pakistan, this issue is made significantly worse by a combination of socioeconomic challenges, a severe shortage of accessible psychiatric professionals, and a persistent cultural stigma that often prevents individuals from recognising their condition or reaching out for help.

Major psychiatric institutions in Lahore, including the Punjab Institute of Mental Health (PIMH) and Fatima Memorial Hospital, consistently report high numbers of patients presenting with depressive disorders across all age groups and economic backgrounds. The volume of patients relative to the number of available mental health professionals highlights both the widespread nature of the problem and the scale of the gap in care. Many individuals with mild to moderate depression remain untreated due to lack of awareness about what depression is, difficulty identifying whether symptoms are serious, financial constraints, geographic barriers, and fear of social judgment. At the same time, the rapid growth of internet access and the use of digital devices across Pakistan presents a meaningful opportunity. Technologies such as artificial intelligence, machine learning, and conversational systems have demonstrated real potential in healthcare, particularly for early screening, patient education, and support. However, their application within Pakistan's specific cultural, linguistic, and healthcare context remains very limited.

This project, NEXORA, was developed in response to this gap. The goal was to build an accessible, private, and stigma-free digital platform that can serve as a first point of contact for someone who may be unsure about their mental health. The system combines a clinically established screening questionnaire (the PHQ-9), a supplementary lifestyle-based risk indicator powered by a machine-learning model, and a supportive chatbot all within a single web application. Importantly, the project was not designed in isolation. The team visited Fatima Memorial Hospital, Fountain house, the Punjab Institute of Mental Health, and Jinnah Hospital, and the guidance received from healthcare professionals at these institutions directly shaped the final design of the system most notably the decision to base the core assessment on the PHQ-9 questionnaire.

1.2 - Problem Statement:

Despite the high prevalence of depression-related distress in Pakistan, a large proportion of affected individuals never engage with any form of structured mental health screening. This problem arises from several interconnected factors.

First, there is a significant lack of awareness among the general public regarding the symptoms and severity of depression. Many individuals interpret their psychological distress as temporary stress or a personal weakness, rather than recognising it as something that may benefit from professional attention. Second, even when individuals acknowledge that something may be wrong, accessing qualified psychiatric professionals is difficult. Geographic barriers, long waiting times in public facilities, and the high cost of private consultations mean that timely care remains out of reach for many people.

Although some digital solutions exist telemedicine platforms, general AI chatbots, and standalone PHQ-9 screening applications they each address only part of the problem. Most AI chatbots do not integrate a standardised clinical assessment tool. Standalone PHQ-9 apps provide a fixed score with no supportive follow-up or supplementary context. Telemedicine platforms focus on scheduling and provide no intelligent first interaction layer.

NEXORA is specifically designed to address the absence of a unified platform that can: (1) perform a structured PHQ-9 screening; (2) provide supplementary lifestyle-risk context; (3) offer supportive conversational guidance; and (4) point users toward professional care all within one accessible web-based system.

1.3 - Project Objectives:

The primary objectives of NEXORA are as follows:

- To implement a structured PHQ-9 screening module with a rule-based severity engine that classifies users into five clinically recognised categories and raises a self-harm/suicide-risk flag when indicated.
- To develop and integrate a supplementary Random Forest machine-learning classifier providing a lifestyle-based risk indication (Low / Moderate / High) from Sleep Quality, Study Pressure, and Financial Pressure. This component clearly complements, and does not replace, the PHQ-9 result.
- To provide a supportive chatbot for psychoeducation and coping guidance, grounded in established psychological literature, without making clinical diagnoses.
- To implement secure user registration and authentication backed by a MySQL database with hashed passwords.
- To deliver a clean, responsive, accessible frontend usable by a non-technical adult user without any training.
- To provide frontend pages for professional guidance and appointment booking.
- To enforce a mandatory authentication gate on the PHQ-9 assessment, and to permanently store each user's assessment results (score, severity level, risk tier, and timestamp) so that a personal assessment history can be reviewed on demand.
- To provide dedicated crisis-support, self-help, and policy pages (Crisis Helpline, Self-Help Articles, Terms of Use, and Privacy Policy) ensuring users have access to emergency resources, wellness guidance, and clear information about platform use and data handling.

1.4 - Scope of Project:

In Scope:

- User registration and login (MySQL-backed, hashed passwords).
- Structured PHQ-9 questionnaire with rule-based scoring and self-harm/suicide-risk flag.
- Mandatory authentication gate for the PHQ-9 assessment; users must be logged in before taking the assessment, and anonymous assessments are not permitted.
- Persistent storage of PHQ-9 assessment results (score, severity level, risk tier, and timestamp) in the MySQL database, linked to the user's account, with historical results viewable on a frontend dashboard/history view.
- Four dedicated guidance and policy pages: Crisis Helpline, Self-Help Articles, Terms of Use, and Privacy Policy, accessible via main navigation and the footer.
- Lifestyle questions (Sleep Quality, Study Pressure, Financial Pressure) feeding a supplementary Random Forest risk classification.
- Flask API layer (/predict and health/status routes) connecting the frontend to the rule engine and the model.
- Supportive chatbot with knowledge-base and fallback.
- Frontend pages: Index, Assessment, Assessment History/Dashboard, Chatbot, Psychologist, Booking, Login/Registration, Crisis Helpline, Self-Help Articles, Terms of Use, Privacy Policy.
- Responsive web design for desktop and mobile browsers.

Out of Scope:

- Clinical diagnosis of depression or any other mental-health disorder.
- Replacement of a psychiatrist, psychologist, doctor, therapist, or emergency crisis service.
- Video or audio consultation capabilities.
- Urdu-language interface in the current version.
- Native mobile application in the current version.

1.5 - Vision Statement:

The vision of NEXORA is to provide a private, stigma-free digital first point of contact for individuals who may be experiencing symptoms of depression combining a validated screening instrument, a supplementary lifestyle-risk indicator, and supportive AI-driven guidance, while at all times clearly distinguishing screening from diagnosis. In its future iterations, the platform envisions expansion to cover additional validated instruments, multilingual support, wearable health data integration, and potential adaptation of the architecture for other disease domains a possibility discussed with healthcare professionals at Jinnah Hospital during the project period.

1.6 - Community Outreach & Social Media Platforms:

To maximize reach, promote mental health literacy, and reduce societal stigma, **NEXORA** extends its platform ecosystem through integrated official social media channels. These channels serve as an external engagement layer to raise awareness, share educational resources, and direct individuals toward the NEXORA web platform for self-assessment, AI chatbot guidance, and professional consultations.

The project incorporates the following official social media and communication handles:

- **Gmail (Official Support & Inquiries):** nexorahealthcare41@gmail.com used for direct user inquiries, administrative communications, platform support, and professional onboarding.
- **Instagram:Page_URL:**https://www.instagram.com/nexora_ahealthcare?stkn=d25pcmM5bmhqZmlo&utm_source=qr /**nexora_healthcare** (Username) dedicated to visual mental health infographics, daily wellness tips, psychoeducation, and platform feature highlights.
- **Facebook:** Page_URL: <https://www.facebook.com/share/1TCU5xjUo2/> **Nexora Healthcare** (Username) for community outreach, hosting educational articles, sharing platform announcements, and building peer support discussions.
- **Threads:** Page_URL: <https://threads.com/share/1TCU5xjUo2/> **nexora_healthcare** (Username) Utilized for real-time announcements, mental health advocacy updates, sharing quick tips, and participating in public health awareness threads.

1.7 - Tools, Libraries, and technologies:

Category	Technology	Purpose & Rationale
Frontend	HTML, CSS, JavaScript, Bootstrap 5	Builds Responsive, interactive user interface
Backend	Python, Flask, Flask-CORS	Handles API management, business logic, user authentication, and rule engine.
Database	MySQL (nexora_db)	Relational database for user registration authentication.
AI / ML Model	scikit-learn Random Forest, joblib	Lifestyle-risk classifier (phq_risk_model.pkl)

Chatbot	Botpress + Gemini API	Provides psychoeducation and coping guidance from a custom knowledge base with Gemini fallback.
IDE / Tools	VS Code, Postman	Primary development environment and API testing tool.
Deployment / Testing	Google Colab, ngrok, GitHub	Backend hosting and API exposure during development/testing

1.8 - System Limitation and Constraints:

- NEXORA is a screening and guidance tool, not a diagnostic instrument, results must never be interpreted as a clinical diagnosis.
- The PHQ-9 rule engine produces a standardised screening result but is influenced by the accuracy and honesty of the user's self-reporting.
- The Random Forest lifestyle-risk model's training data, methodology, and performance metrics require full documentation and are not yet published within this document.
- The Google Colab and ngrok development arrangement is not a production-grade deployment. Colab sessions time out and ngrok URLs change between sessions.
- The system is currently available in English only.
- A native mobile application is not included in the current version.

Chapter No. 2
LITERATURE REVIEW

2.1 - Overview:

This chapter reviews existing systems, tools, and relevant academic research in the area of mental health screening, conversational support agents, and digital mental health platforms. The purpose is to understand the current state of the field, identify the limitations and gaps in existing approaches, and justify the design decisions made in developing NEXORA. The review covers the most widely studied platforms in this domain, as well as the academic evidence supporting the use of standardised tools like the PHQ-9.

2.2 - Review of Existing Systems:

2.2.1 Woebot Health

One of the most widely studied AI-based mental health chatbots is Woebot, which is based on the principles of Cognitive Behavioural Therapy. It engages users through structured therapeutic conversations and incorporates features such as mood tracking. Research published in the Journal of Medical Internet Research has demonstrated its effectiveness in reducing symptoms of depression and anxiety, particularly among young adults over short-term usage. Despite its strengths, Woebot has certain limitations in the context of this project. It primarily functions as a standalone conversational agent and does not provide integration with local mental health professionals for further clinical support. Additionally, it is not directly linked with a standardized clinical assessment tool for evaluating depression severity. Its deployment is also largely focused on English-speaking populations, with limited adaptation to diverse cultural and regional contexts such as Pakistan.

These limitations highlight the need for a more integrated and locally relevant solution that combines clinical assessment, AI-based guidance, and access to professional mental health services within a single platform.

2.2.2 Wysa

Wysa is an AI emotional support chatbot that uses natural language processing to provide CBT, Dialectical Behaviour Therapy (DBT), and mindfulness-based exercises. It has demonstrated clinical effectiveness in peer-reviewed studies and has been adopted in several healthcare systems. However, much like Woebot, Wysa does not integrate a validated clinical scoring mechanism such as the PHQ-9, nor does it facilitate real-world appointment booking with local psychiatrists. The transition from AI-guided support to professional care requires the user to independently seek out a healthcare provider, representing a critical gap in the care pathway.

2.2.3 Telemedicine Platforms (Marham.pk, Sehat Kahani)

Pakistan-based telemedicine platforms such as Marham.pk and Sehat Kahani provide valuable appointment booking services that connect patients with doctors, including

psychiatrists. However, these platforms function primarily as directory and scheduling services; they lack any AI-powered triage or assessment layer. A user arriving at these platforms must already possess an awareness of their condition and a decision to seek professional help. There is no integrated screening mechanism to guide users who may be uncertain about the severity of their mental health condition, nor is there any initial psychoeducational support offered prior to the appointment.

2.2.4 PHQ-9 Based Screening Applications

Several standalone web and mobile applications implement the Patient Health Questionnaire-9 (PHQ-9) for depression screening. The PHQ-9 is a clinically validated, nine-item instrument widely used by healthcare professionals for detecting and monitoring depression severity. However, these implementations are predominantly static; they present a fixed-scoring rubric and display a result category without any personalised follow-up action. They do not integrate machine learning for nuanced classification, do not provide subsequent therapeutic guidance, and offer no connection to professional healthcare services.

2.2.5 General-Purpose AI Chatbots (ChatGPT, Gemini)

Large language models such as OpenAI's ChatGPT and Google's Gemini have been explored informally for mental health support. While these models possess vast general knowledge and can engage empathetically, they present significant concerns for clinical use. They are not specifically trained on psychiatric datasets, they lack a structured assessment framework, and their outputs are neither deterministic nor consistently clinically appropriate. They also cannot verify the credentials of users or connect them with professional services.

2.3 - Gap Analysis:

The review of existing mental health assessment and guidance systems reveals several important gaps that the proposed AI-Based Mental Health Risk Prediction and Guidance System (**NEXORA**) aims to address. Although existing platforms provide useful individual services such as mental health questionnaires, chatbots, or access to professionals, these services are often separated and do not provide a complete and structured user journey. The proposed system addresses these limitations by combining validated assessment, supplementary lifestyle-based risk analysis, AI-assisted guidance, user authentication, and professional support within a single web-based platform.

1. Lack of Integrated Mental Health Assessment and Guidance:

Many existing mental health platforms focus on only one aspect of mental health support, such as questionnaire-based screening or chatbot-based assistance. Users may therefore need to use different platforms for assessment, guidance, and professional support. The proposed system integrates the **PHQ-9 questionnaire** with an additional **AI-based lifestyle risk assessment** and

an AI-assisted chatbot within one platform. This provides users with a more organized and convenient experience while keeping the clinically recognized PHQ-9 assessment as an important part of the system.

2. **Limited Use of Supplementary Lifestyle Factors:**

Mental health risk cannot always be understood through a single questionnaire score. Lifestyle-related factors such as sleep quality, study pressure, and financial pressure may also provide useful contextual information about a user's overall well-being. To address this limitation, the proposed system includes a supplementary lifestyle-based assessment. The selected lifestyle responses **Good, Average, Bad, and Worst** are converted into numerical values and processed by the machine learning component to generate an additional risk indication. This result is presented separately from the PHQ-9 result so that the supplementary lifestyle assessment does not replace or misrepresent the validated PHQ-9 screening.

3. **Lack of Safe and Knowledge-Grounded AI Guidance:**

General-purpose AI chatbots can generate responses that may not always be appropriate for sensitive mental health questions. The proposed system therefore uses a **Botpress-based chatbot with a psychology-focused knowledge base**. The knowledge base is derived from *Abnormal Psychology* by **Ronald J. Comer and Jonathan S. Comer** and is intended to provide initial psychoeducation, general coping strategies, and empathetic support. When a user's question is outside the intended knowledge-base scope, the chatbot can use its configured AI fallback mechanism to provide a response. This approach provides a more structured form of conversational assistance rather than relying entirely on unrestricted AI-generated responses.

4. **Absence of a Complete User and Professional Support Pathway:**

Some existing systems provide mental health screening or conversational assistance but do not connect these functions with other supporting services. The proposed system extends beyond assessment by providing **user registration and login through a MySQL database**, mental health assessment, result presentation, chatbot assistance, and access to psychiatrist-related functionality, including the booking component. This creates a more complete pathway from user access and assessment to guidance and potential professional consultation.

5. **Limited Emphasis on Responsible Use of AI in Mental Health:**

A major concern with AI-based mental health systems is the possibility of presenting AI output as a medical diagnosis. The proposed system is designed as a **risk prediction and guidance system rather than a diagnostic tool**. The PHQ-9 is used as a standardized screening instrument, while the lifestyle-based assessment is presented as supplementary information. The chatbot provides general psychoeducation and supportive guidance and is not intended to replace a qualified psychologist, psychiatrist, or other healthcare professional. Users with concerning results can therefore be encouraged to seek appropriate professional support rather than relying solely on the system.

6. **Lack of a Locally Relevant Development and Validation Perspective:**

Mental health applications should consider the needs and context of the population for which

they are developed. During the development of this project, the team sought guidance from healthcare professionals and hospital management at **Fatima Memorial Hospital, the Punjab Institute of Mental Health (PIMH), and Jinnah Hospital**. Discussions and meetings with doctors, psychologists, hospital management, and relevant departments provided practical guidance regarding mental health, psychological assessment, patient safety, and the appropriate use of screening questionnaires. This professional input helped the team refine the system and strengthen the role of PHQ-9 as a safer and more structured component of the assessment process.

Summary of the Gap

The proposed system addresses these gaps by combining **PHQ-9-based screening, supplementary lifestyle-based risk prediction, knowledge-base-grounded chatbot assistance, AI fallback support, MySQL-based user authentication, and professional support functionality** in a single web application. Rather than depending entirely on an AI chatbot, the system uses a structured assessment process and separates the clinically recognized PHQ-9 result from the supplementary lifestyle risk indication. This design aims to provide users with accessible, understandable, and responsible preliminary mental health guidance while maintaining the importance of professional healthcare consultation.

Chapter No. 3
SYSTEM ANALYSIS

3.1 - Overview:

This chapter provides a thorough analysis of NEXORA from the perspectives of feasibility, functional requirements, non-functional requirements, risk, and strategic positioning. It also documents a critically important element of our development process: our direct interactions with healthcare institutions and medical professionals, which significantly influenced the requirements and design of the system.

3.2 - Feasibility Study:

3.2.1 Technical Feasibility

NEXORA is built entirely on well-established, open-source technologies. Python and Flask provide a reliable and lightweight backend framework. MySQL is an industry-standard relational database with a long track record of stability. The scikit-learn library provides a reliable Random Forest implementation. Bootstrap 5 ensures a responsive, cross-platform-compatible user interface. Google Colab provides a free, cloud-based Python environment suitable for running the Flask backend and machine-learning model during development and testing. All technologies are freely available, and the development team has the necessary expertise in each area.

3.2.2 Economic Feasibility

The project is economically viable with minimal financial outlay. All core development technologies are open-source and free of charge. Google Colab and MySQL Community Server are free. Deployment to a production environment would require a modest cost for cloud hosting, but free or low-cost tiers are available on platforms such as Render or Railway for academic-scale applications.

3.2.3 Operational Feasibility

The system is designed with a non-technical end user as the primary persona. The PHQ-9 questionnaire is presented in a clean, linear format with simple four-option response choices. The lifestyle questions use the same familiar format. The chatbot requires no technical knowledge. The system is operationally feasible for adoption by the intended user base Pakistani adults who may be experiencing symptoms of depression.

3.3 - Functional Requirements:

FR ID	Requirement Name	Description
FR01	User Registration	Allows new users to create an account by providing their full name, email address, and password. Credentials are stored in the MySQL database with passwords hashed. Email must be unique.
FR02	User Login	Allows registered users to securely log in using email and password. Backend verifies password against stored hash.
FR03	PHQ-9 Questionnaire	Presents authenticated users with a structured nine-item PHQ-9 questionnaire; the assessment is accessible only after the user has logged in, and anonymous submissions are not permitted. Each question offers four response options scored from 0 to 3. The total score is computed upon submission.
FR04	PHQ-9 Severity Classification	Rule engine classifies total score: Minimal (0–4), Mild (5–9), Moderate (10–14), Moderately Severe (15–19), Severe (20–27).
FR05	Self-Harm/Suicide-Risk Flag	If PHQ-9 item 9 response > 0, a risk flag is raised and displayed with crisis guidance.
FR06	Lifestyle-Factor Questions	Three lifestyle questions: Sleep Quality, Study Pressure, Financial Pressure. Options: Good, Average, Bad, Worst.
FR07	Lifestyle-Risk Classification	Frontend encodes selections (0–3); Flask passes values to Random Forest model; returns Low/Moderate/High classification.
FR08	Results Display	Results page shows PHQ-9 score, severity label, self-harm flag, and supplementary lifestyle-risk classification with clear labels and guidance. The result is permanently saved to the user's account and becomes available in the assessment history view.
FR09	Chatbot	Integrates a Botpress chatbot that greets the user, queries its custom knowledge base for relevant responses, and falls back seamlessly to the Gemini API for queries irrelevant or outside the knowledge base, all within the same interface.

FR-10	Professional Guidance Page	Displays a list of psychologists including their Name, Profile Picture, Specialization, and Contact Information.
FR-11	Booking Page	Frontend page for exploring and scheduling appointment booking.
FR-12	User Logout	Allows the user to securely terminate their active session.
FR-13	Persistent Assessment Storage & History	Each completed PHQ-9 assessment (score, severity level, risk tier, and exact timestamp) is permanently stored in the MySQL database (phq9_assessments table), linked to the authenticated user's account via user_id. Users can view their historical assessment results directly on the frontend dashboard/history view.
FR-14	Guidance & Policy Pages	Provides four dedicated static pages, accessible via main navigation and the footer: a Crisis Helpline page listing local 24/7 emergency contacts; a Self-Help Articles page with categorised wellness resources; a Terms of Use page; and a Privacy Policy page.

3.4 - Non-Functional Requirements:

NFR ID	Attribute	Description
NFR-01	Data Privacy	All user data, Passwords stored as hashes, must be stored securely using bcrypt. The system must not expose personal health information, including stored PHQ-9 assessment records, to any third party.
NFR-02	Performance	The questionnaire submission and classification result must be returned within 3 seconds under normal network conditions. The chatbot must respond within 5 seconds.
NFR-03	Usability	The interface must be intuitive and navigable by a non-technical adult user without any training or guidance, following established web UX conventions.
NFR-04	Reliability	The appointment booking module must ensure, a slot marked as booked must never be concurrently assigned to two different users.
NFR-05	Scalability	The system architecture must support future feature additions, including additional assessment instruments and expanded doctor profiles, without requiring a complete redesign.

NFR-06	Compatibility	The web application must be fully functional on all major modern browsers (Chrome, Firefox, Edge, Safari) and must be responsive on mobile screen sizes.
NFR-07	Maintainability	The codebase must be modular and well-commented to facilitate future debugging, enhancement, and handover.

3.5 - SWOT Analysis:

Strengths	Weaknesses
Combines PHQ-9 with supplementary lifestyle-risk indicator. Informed by professional guidance from three major hospitals. PHQ-9 adoption based on direct clinical recommendation. Open-source, low-cost stack. Supportive chatbot with knowledge base.	Random Forest training data and performance metrics not yet fully documented. Colab/ngrok is not production-grade. No Urdu interface. Booking/guidance backend status requires confirmation.
Opportunities	Threats
Growing mental health awareness in Pakistan. Additional instruments possible (GAD-7, PCL-5). Expressed interest from Jinnah Hospital in expansion to other disease domains. Mobile app development potential.	Mental health stigma discouraging use. Misinterpretation of screening results as clinical diagnosis. Colab/ngrok session instability.

3.6 - Risk Analysis:

Risk	Likelihood	Impact	Mitigation Strategy
Misinterpretation of results as diagnosis	Medium	High	Prominent disclaimers on all results pages and in chatbot responses.

Colab/ngrok session interruption	Medium	Medium	Plan stable hosting before production or evaluation deployment.
User data privacy exposure	Low	High	Hash all passwords; use environment variables; parameterised SQL queries.
Random Forest model underperforming	Unconfirmed	Medium	Document and validate training data and performance metrics; present as supplementary indicator only.

3.7 - Domain Validation and Clinical Guidance /Hospital Interactions:

A distinctive and important aspect of NEXORA's development is that it was not built purely as a programming exercise. The team made a deliberate effort to engage with healthcare professionals and institutions in order to understand mental health from a clinical and operational standpoint, validate the proposed system design, and ensure that the platform is built responsibly and ethically. The following subsections describe these interactions.

3.7.1 Professional Consultation by Prof. Dr. Syed Waqar ul Qounain

During the initial phase of the Final Year Project, our team visited the **University of the Punjab** and met **Prof. Dr. Syed Waqar ul Qounain**, Director of the Artificial Intelligence and Multidisciplinary (AIM) Research Lab and Professor/Chairman of the Department of Information Technology, in his office. We discussed our initial project ideas and sought his guidance regarding a suitable and meaningful problem to address. During the discussion, he highlighted the growing importance of **mental health**, particularly the increasing prevalence of issues such as stress, anxiety, and depression in society, and suggested that we consider developing a technology-based solution in this area. This discussion became an important starting point for our project, and our team subsequently began working on the **NEXORA, A Mental Health Risk Prediction and Guidance System**. To strengthen the project with practical and domain-specific knowledge, we conducted extensive research and later visited several hospitals, mental health institutions, and private psychological clinics across Lahore, including **Fountain House, Mayo Hospital, PIMH, Fatima Memorial Hospital, and Jinnah Hospital**.

3.7.2 Professional Guidance from Dr. Aqsa Naeem (Lecturer)

Dr. Aqsa Naeem provided valuable guidance to our team throughout the development of the project. She helped us understand appropriate ways of communicating with individuals experiencing mental health difficulties and guided us regarding empathetic interaction, patient observation, and responsible mental health support. She also contributed to the development of our chatbot's knowledge base by recommending *Abnormal Psychology* by Ronald J. Comer and Jonathan S. Comer as an important academic reference covering major psychological disorders and related areas such as depression, anxiety, and stress. Through her guidance and our field visits, we had opportunities to observe and interact with individuals experiencing psychological difficulties. These experiences helped us understand how individuals communicate their concerns and how factors such as facial expressions, body language, tone, and general behaviour can provide important contextual information during an interaction. We also learned the importance of listening without judgment, maintaining confidentiality, showing empathy, and creating a comfortable environment for individuals to express their concerns. Her continued guidance significantly contributed to our understanding of the psychological aspects of the project and helped us develop NEXORA with a stronger focus on responsible, supportive, and human-centred mental health guidance.

3.7.3 Fatima Memorial Hospital

The team visited Fatima Memorial Hospital, one of the major healthcare institutions in Lahore, early in the project development period. We interacted with hospital personnel through the Ethics Review Board (ERB) department. The Chairperson, Shahima Rehman, and her Personal Assistant provided guidance related to our project and ERB department.

The guidance received at Fatima Memorial Hospital was particularly important in helping the team understand the ethical dimensions of developing a digital mental health system. Key areas of guidance included:

- Ethical considerations in collecting and handling sensitive personal and health-related information.
- The importance of clearly communicating the limitations of a digital screening tool to users including that the system must not claim to diagnose any condition.
- Appropriate handling of sensitive user data, including the need for secure storage and privacy protection.
- The requirement for any mental health-related digital tool to guide users toward professional help rather than substituting for it.
- Guidance on the responsible and ethical design of a system dealing with a topic as sensitive as mental health and depression.



Fatima Memorial Hospital

3.7.4 Punjab Institute of Mental Health (PIMH)

The team also visited the Punjab Institute of Mental Health (PIMH), one of Pakistan's foremost dedicated psychiatric institutions. During this visit, we interacted with doctors, psychologists, and administrative personnel who deal with mental health assessment and patient care on a daily basis.

The interaction with PIMH professionals deepened our understanding of:

- How mental health assessments are conducted in a professional clinical setting.
- The range of psychological symptoms associated with depression and how they manifest across different patients.
- The types of concerns and questions that individuals with mental health conditions typically have when they first seek help.
- How a digital system should communicate risk information to a lay user without causing undue alarm or misinterpretation.

- The importance of empathy and non-judgmental language in all mental health communications.

These insights directly influenced how NEXORA presents its results and chatbot responses with careful, supportive language and prominent reminders that the system is a screening tool and not a substitute for professional evaluation.





Department of Psychiatry

3.7.5 Jinnah Hospital, Lahore

The team's interaction with Jinnah Hospital was particularly significant and directly shaped the core design of the NEXORA assessment system. We met with the Medical Superintendent and several Heads of Departments. These professionals reviewed our project proposal and provided substantive professional guidance.

The Medical Superintendent and Heads of Departments expressed genuine appreciation for the concept and the goal of making mental health screening more accessible. More importantly, they provided specific technical and clinical guidance that changed the final design. In particular, they strongly recommended incorporating the Patient Health Questionnaire-9 (PHQ-9) as the primary structured screening instrument.

Their reasoning for recommending the PHQ-9 was well-founded:

- The PHQ-9 is a widely validated, clinically established tool used by mental health professionals worldwide, including in Pakistan.

- Using a standardised instrument like the PHQ-9 is a safer and more responsible approach than relying entirely on an AI model or chatbot to assess a person's mental health.
- A structured questionnaire produces consistent, interpretable results that can be meaningfully related to clinical severity categories.
- Combining a structured clinical questionnaire with supplementary AI-based lifestyle factors is more defensible and reliable than using AI alone.

This professional feedback was a defining moment in the project. Based on the guidance received, the team made the deliberate design decision to anchor the core assessment in the PHQ-9, with the machine-learning component serving only as a supplementary, clearly-labelled indicator. This approach structured clinical questionnaire + supplementary AI/lifestyle indicator directly reflects the professional advice from Jinnah Hospital.





Department of Psychiatry

3.7.6 Hospital Meetings, Project Proposal, and Review from Clinical and Administrative Staff

Across all three institutions, the team attended several meetings in which the project proposal was presented, discussed, and evaluated by healthcare professionals. These meetings served several important purposes:

- They allowed us to present the concept of a digital mental health screening and guidance system to professionals who work in the field every day, and to receive their honest assessment.
- They helped us understand what healthcare professionals consider essential, appropriate, and potentially problematic in such a system.
- They provided domain-specific knowledge about mental health assessment that could not have been acquired purely through academic reading.

- They confirmed that there is genuine professional interest in and support for technology-based tools that can help bridge the gap between patients and mental health services, provided they are designed responsibly.

The feedback received during these meetings had several concrete effects on the project:

- The PHQ-9 was incorporated as the primary structured assessment instrument based on direct professional recommendation from Jinnah Hospital.
- The system was designed to clearly and consistently distinguish between screening and diagnosis in all interfaces.
- Chatbot responses were designed to be supportive and educational rather than clinical.
- The results page was designed to guide users toward professional consultation when scores indicate elevated severity.

These interactions demonstrate that NEXORA was developed not just as a technical exercise, but as a project genuinely informed by the healthcare professionals who understand its real-world implications.

3.7.7 Project Discussion and Guidance from Healthcare Professionals at Fatima Memorial Hospital

During the development of the project, our team visited **Fatima Memorial Hospital** to discuss the proposed mental health system with healthcare professionals and gain a better understanding of psychological and psychiatric considerations. We discussed the project with **Dr. Saima Batool**, **Dr. Junaid Rasool (Head of the Psychiatry Department)**, **Chairperson Shahima Rehman**, and **Dr. Muhammad Farhan Feroz (MS)**, along with the Chairperson's PA. Their guidance helped us understand the ethical considerations involved in developing a mental health system, particularly the importance of considering patients' questions, concerns, and safety. We also interacted with students to gain additional perspectives and improve our understanding of mental health-related requirements. These discussions helped our team approach the project from both a technical and human-centered perspective.



Mrs. Shahima Rehman (The chairperson of Fatima Memorial Hospital and the NUR-Fatima Memorial System)



Dr Junaaid Rasool (Head of Psychiatry Department)
Psychiatrist)



Dr Saima Batool (Consultant

3.7.8 Project Discussion and Guidance from the Punjab Institute of Mental Health (PIMH)

Our team also visited the **Punjab Institute of Mental Health (PIMH)**, where we interacted with doctors and psychologists and arranged a panel meeting to formally present our proposed system. During the meeting, we explained the objectives and working concept of the project, after which the panel provided guidance about how mental health services and patient-related processes work in practice. We discussed the project in detail with **Prof. Dr. Aysha Rashid (MS)**, **Dr. Emaan** (+92 330-4242123), **Dr. Alina** (Contact: +92 336-3340001), **Ms. Aqila Unbrin (Principal Clinical Psychologist)**, Contact: +92 306-9286011), and **Ms. Saima Majeed**. One of the concerns raised during the discussion was that an AI-based system should not be designed in a way that replaces healthcare professionals. The panel also emphasized the importance of addressing patients with severe mental health conditions appropriately. Based on this professional feedback, our team further refined the system so that AI would serve as a **supportive tool rather than a replacement for doctors or psychologists**, while also providing an appropriate pathway for users who may require professional attention.



Ms. Aqila Unbrin (Principal Clinical Psychologist)



Ms. Rabia Faiz (Clinical Psychologist)



Ms. Saima Majeed (Senior Clinical Psychologist)



Ms. Alina (Senior Clinical Psychologist)



3.7.9 Project Discussion and Guidance from Healthcare Professionals at Jinnah Hospital

Our team visited **Jinnah Hospital** several times to discuss the project with healthcare professionals and hospital management. Before further development, we obtained approval from **Dr. Syed Mohsin Ali Shah (MS, Jinnah Hospital)**. We subsequently discussed the project with **Dr. Saida Yasir (HOD, Contact: 0335-9996824)**, who appreciated the proposed work and offered the opportunity to develop a similar system for use in the hospital and potentially extend it to other diseases and medical fields. We were also in contact with **Dr. Tehreem (Psychiatry Ward, PA)** and **Dr. Mohsin (Psychiatry Ward, Contact: 0303-708058)**, who guided us regarding hospital procedures, data collection considerations, and the process of preparing and submitting proposals and documents to the **ERB department**. We also interacted with **Dr. Mehwish, Mr. Bashir (Contact: 0333-4285699)**, **Ms. Saira (Contact: 0333-8694032) from the Community Department**, and **Dr. Harish (Contact: 0331-6546129)**, who provided additional guidance and shared their perspectives on the practical working of the hospital system.



Dr Saida Yasir (Head of Psychiatry Department)



3.7.10 Project Discussion with HOD at Jinnah Hospital

Jinnah Hospital, we discussed our project with Dr. Saida Yasir (Head of Psychiatry Department). She highly appreciated our work and showed great interest in the proposed system.

She expressed that, after the successful completion of the project, she would like our team to provide the system to the hospital and further develop similar solutions for other diseases and medical fields. Her appreciation and interest encouraged us to consider the future expansion of the project beyond mental health.

- Furthermore, **Dr. Mohsin Ali** descry our whole system and facilitate us in the peregrination of collection of data or ERB document submission.



Dr. Mohsin Ali

CHAPTER NO. 4
SYSTEM DESIGN AND METHODOLOGY

4.1 - Overview:

This chapter presents the complete architectural and workflow design of NEXORA. It describes the three-tier system architecture, the step-by-step workflows for assessment and authentication, the UML diagrams, and the database design. All design decisions documented here are directly traceable to the functional and non-functional requirements outlined in Chapter 3, and reflect the actual implemented system.

4.2 - System Architecture:

NEXORA follows a classic three-tier client-server architecture, comprising a Presentation Tier, an Application Tier, and a Data Tier. This separation of concerns ensures that each layer can be developed, maintained, and updated independently.

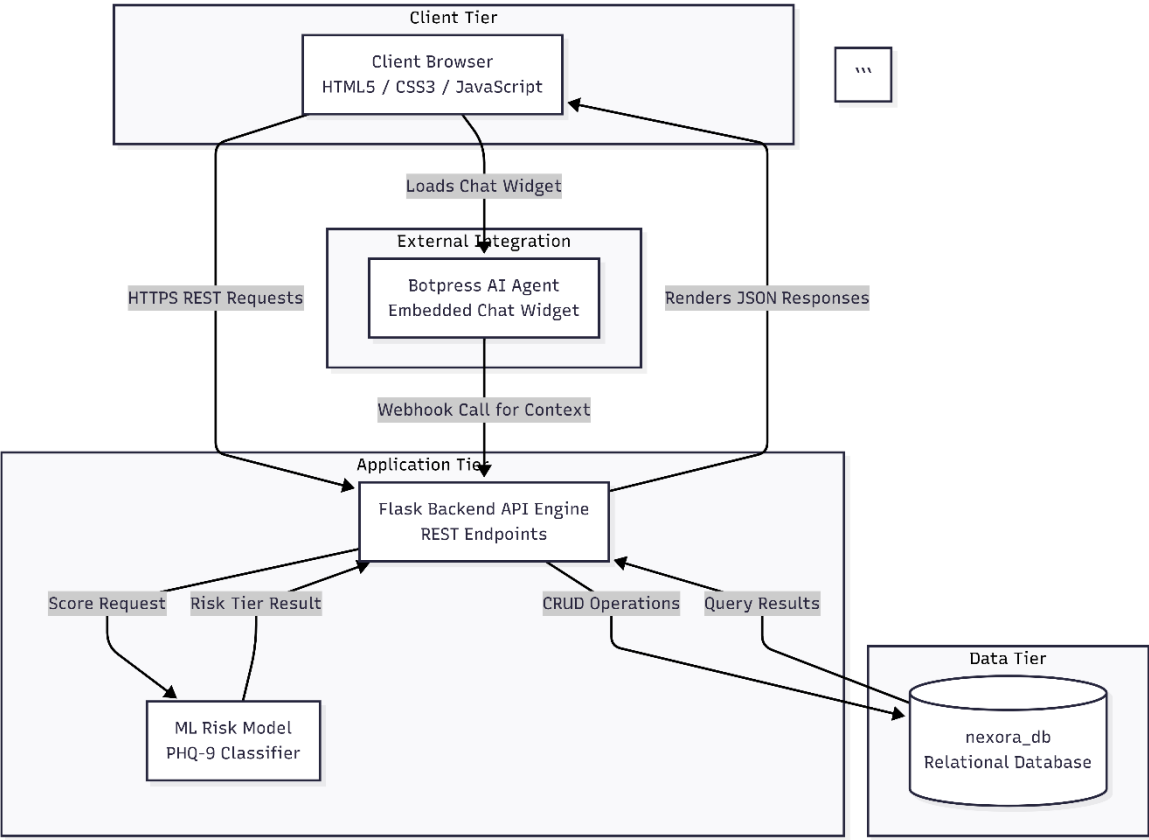
Presentation Tier: The frontend is developed using HTML5, CSS3, JavaScript (ES6), and Bootstrap 5. This layer renders all web pages, manages user interactions, and communicates with the Application Tier by making HTTP requests to Flask API endpoints. JavaScript handles form collection, input encoding, asynchronous API calls via the Fetch API, and dynamic display of results.

Application Tier: The Python Flask application forms the core of the business logic layer. It receives requests from the frontend, validates inputs, runs the PHQ-9 rule engine, loads the serialised Random Forest model (phq_risk_model.pkl), constructs a combined JSON response, and manages user authentication against the database.

Data Tier: The MySQL relational database (nexora_db) serves as the persistent storage layer, housing user registration and authentication records in the users table, and permanently storing each completed PHQ-9 assessment (score, severity level, risk tier, and exact timestamp) in the phq9_assessments table, linked to the owning user via a user_id foreign key.

During development and testing, the Flask application ran inside a Google Colab notebook, and ngrok was used to expose the Colab-hosted Flask server through a public HTTPS URL. This is a development and testing arrangement, not a permanent production deployment. The overall communication flow is:

User (Browser) → HTML/JS/Bootstrap → HTTP POST → Flask API (Colab+ngrok) → Rule Engine + Random Forest → JSON Response → Frontend Display



4.3 - System Workflow:

4.3.1 Assessment and Prediction Workflow

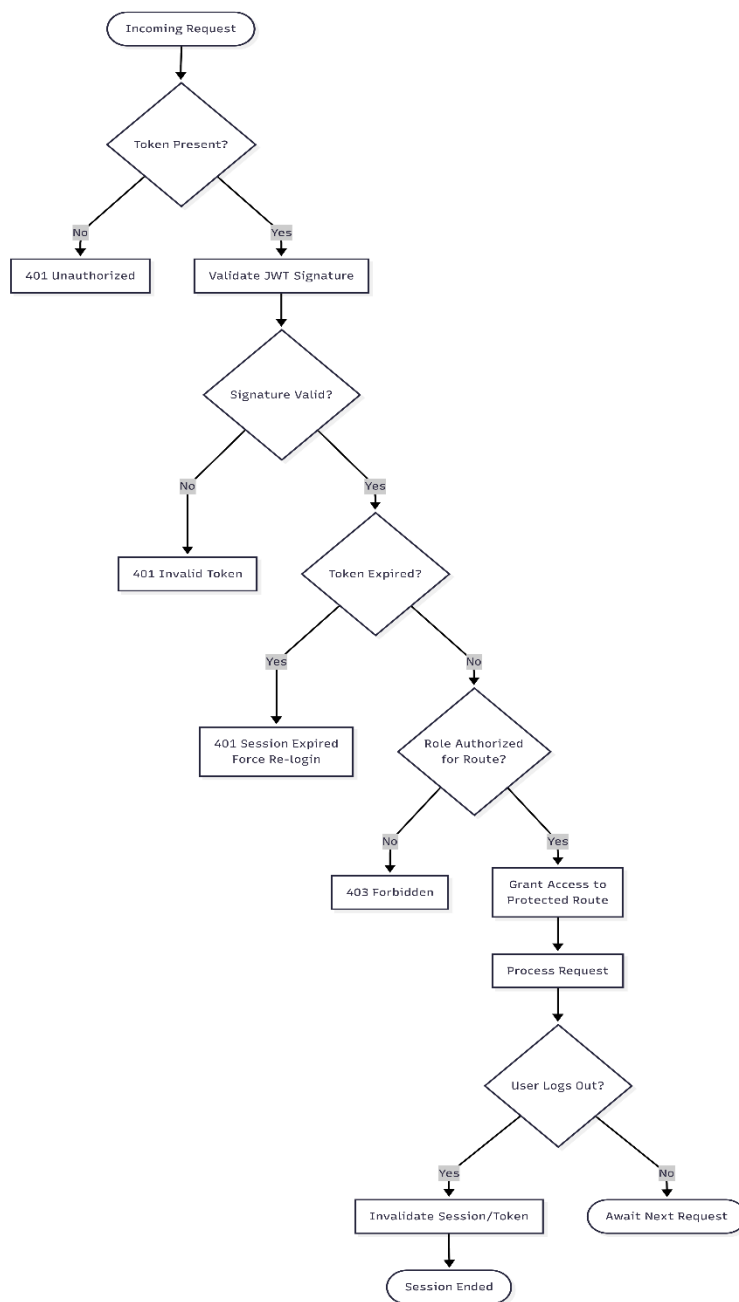
Steps	Actor/ Component	Action
1	User	Logs in, then opens the assessment page (access is blocked for unauthenticated users); answers 9 PHQ-9 questions and 3 lifestyle questions.
2	Frontend (JS)	Collects all responses. Converts lifestyle selections to numbers: Good=0, Average=1, Bad=2, Worst=3.

3	Frontend (JS)	Sends POST request to /predict with 9 PHQ-9 answers, sleep_quality, study_pressure, financial_pressure as JSON.
4	Flask API	Receives and validates the JSON request.
5	Flask/ Rule Engine	Sums the 9 PHQ-9 scores; applies rule engine to determine severity category; checks item 9 for self-harm flag.
6	Flask API	Passes the three lifestyle numeric values to phq_risk_model.pkl (Random Forest).
7	Random Forest Model	Returns lifestyle-risk classification (Low/Moderate/High) and class probabilities
8	Flask API	Returns combined JSON: PHQ-9 score, severity label, self-harm flag status, lifestyle-risk classification.
9	Frontend (JS)	Renders results page showing PHQ-9 result and lifestyle-risk result with clear labels, guidance text, and disclaimer.

4.3.2 Authentication Workflow:

Registration: User fills name, email, password on registration page. Frontend sends data to Flask backend. Backend hashes the password and inserts a new record into the users table in nexora_db. If email already exists, appropriate error is returned.

Login: User submits email and password. Flask retrieves the matching user record from the users table, verifies the submitted password against the stored hash, and returns an authentication result to the frontend.



4.3.3 Chatbot Workflow

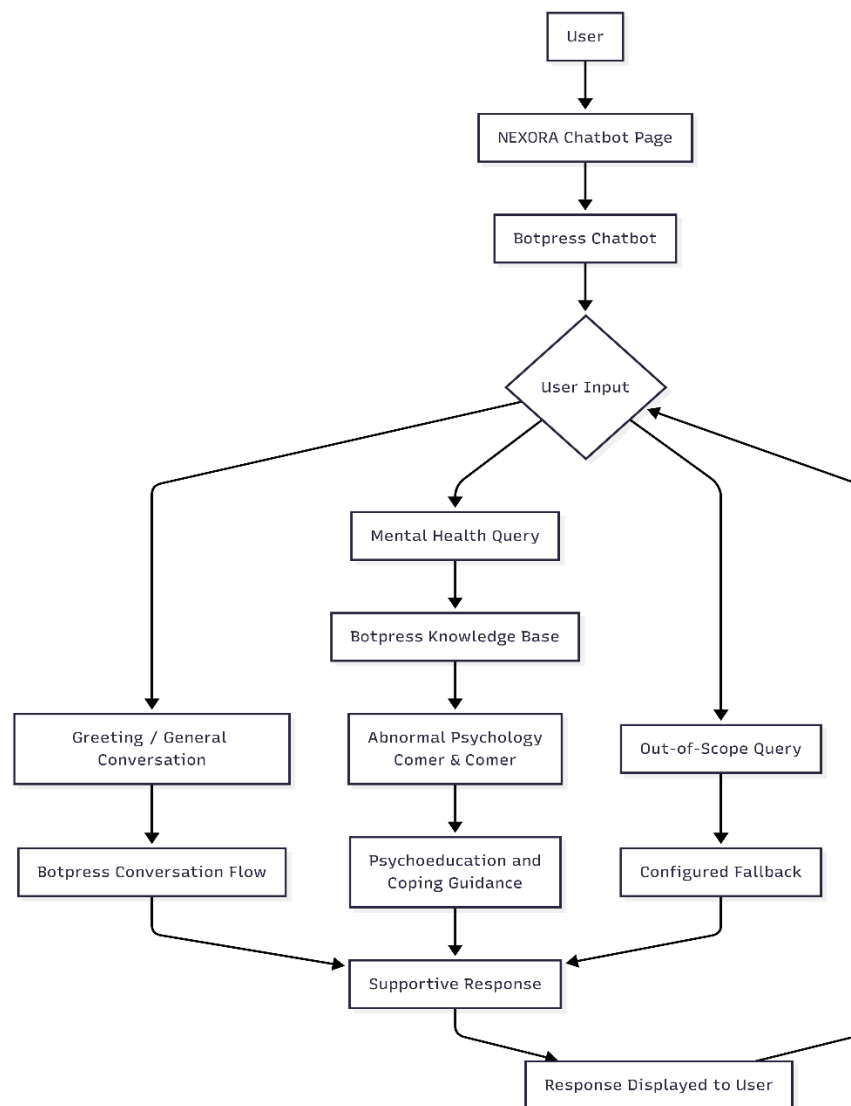
Step 1: User asks a question: User types a message into the chatbot interface.

Step 2: Knowledge-base lookup: The chatbot searches its psychology knowledge base (derived from Abnormal Psychology by Comer & Comer) for a relevant response.

Step 3: Knowledge-base response: If the query matches the knowledge base with sufficient confidence, the response is generated from it. This covers psychoeducation, mental health information, coping strategies, and supportive responses.

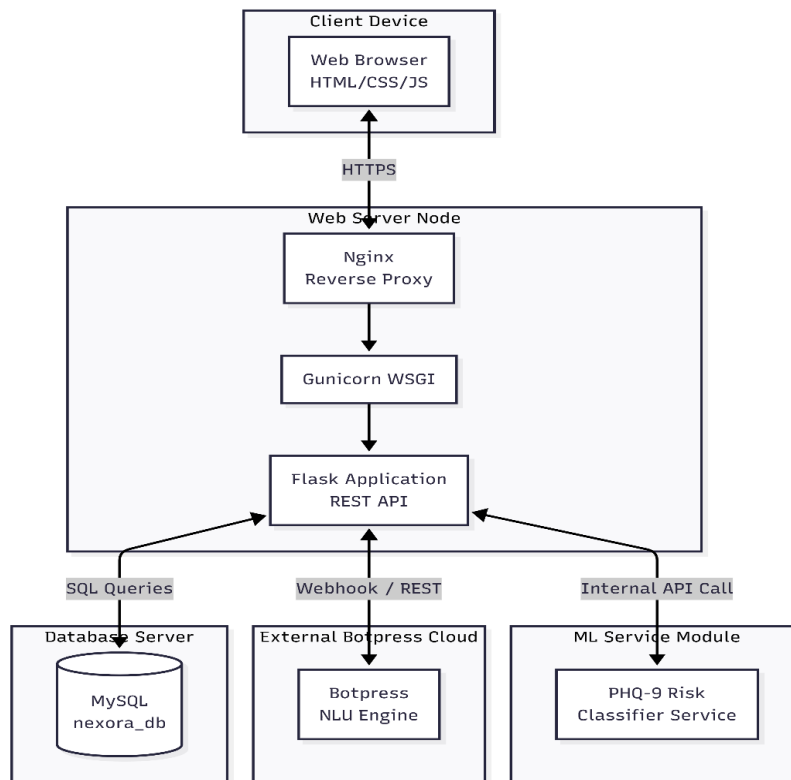
Step 4: Fallback for out-of-scope queries: If the query falls outside the knowledge base or is unrelated to mental health, the system uses a fallback external AI service to handle it gracefully.

Step 5: Response delivery: The chatbot presents its response. All responses are supportive and educational, not clinical. Crisis resources are provided when distress indicators are present.



4.4 – UML Diagrams:

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user identifier, automatically assigned.
full_name	VARCHAR(100)	NOT NULL	User's full name as entered during registration.
email	VARCHAR(150)	NOT NULL, UNIQUE	User's email address. UNIQUE constraint prevents duplicate registrations.
password_hash	VARCHAR(225)	NOT NULL	User's password stored as a cryptographic hash. Never stored as plain text.
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Timestamp recording when the user account was created.



- **PHQ9 assessments Table:**

This table was added to support the mandatory authentication gate on the PHQ-9 assessment: it permanently records each completed assessment against the submitting user, enabling the frontend history/dashboard view.

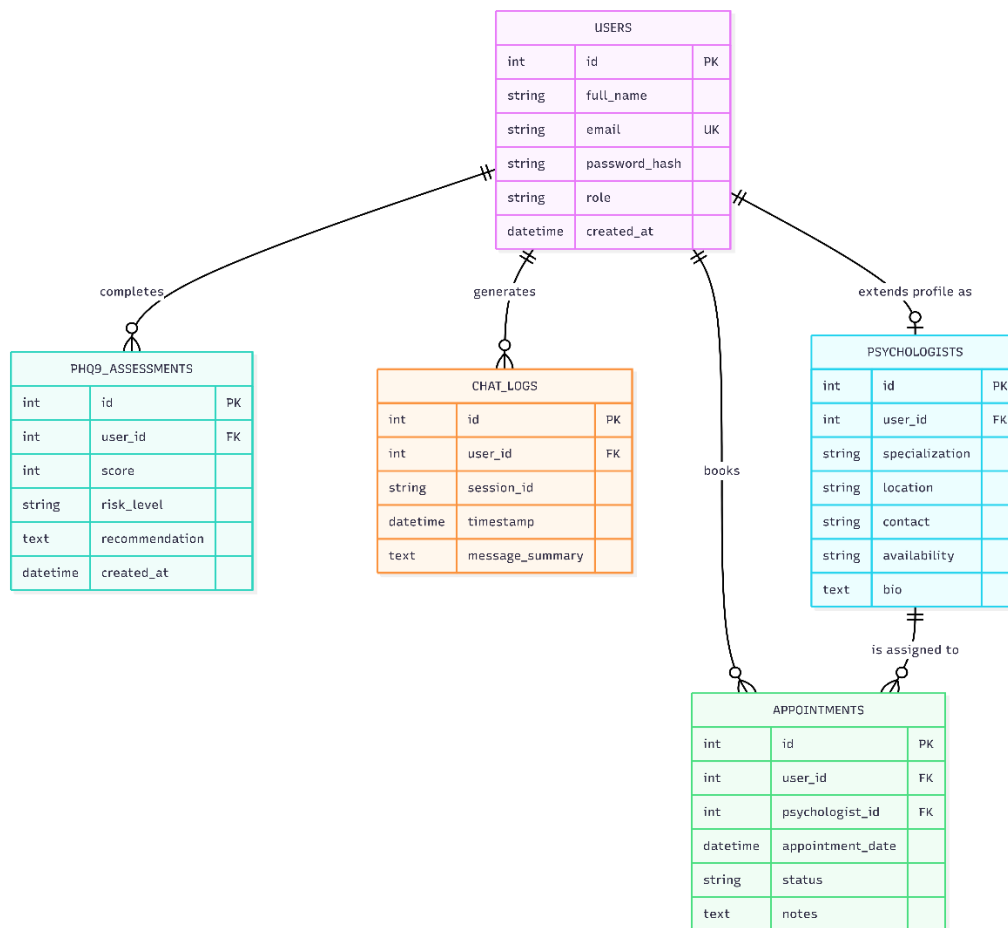
Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the assessment record, automatically assigned.
user_id	INT	NOT NULL, FOREIGN KEY (users.id)	References the authenticated user who completed the assessment.
phq9_score	INT	NOT NULL	The total PHQ-9 score computed at submission time (range 0-27).
severity_level	VARCHAR(30)	NOT NULL	The PHQ-9 severity category (Minimal, Mild, Moderate, Moderately Severe, Severe).
risk_tier	VARCHAR(20)	NOT NULL	The supplementary lifestyle-risk classification (Low, Moderate, High) returned by the Random Forest model.
assessment_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Exact timestamp recording when the assessment was submitted and stored.

The user_id column establishes a foreign-key relationship to the users table, so every stored assessment is uniquely linked to the account that submitted it. The iterative methodology was chosen for its flexibility and ability to incorporate continuous feedback, which is crucial for developing a user-centred application like a healthcare chatbot. This approach allowed for ongoing improvements based on user interactions and feedback, ensuring a robust and effective final product.

4.5 - Database Design:

The MySQL database is named nexora_db. The following tables are confirmed for the current implementation: the users table, which stores account and authentication records, and the phq9_assessments table, which permanently stores each user's PHQ-9 score, severity level, risk tier, and submission timestamp.

Why a database is required: Without a database, the system would have no way to remember registered users, verify login credentials, protect user accounts between sessions, or retain a user's past PHQ-9 results for later viewing. MySQL was selected because it is a well-established, reliable, and open-source relational database management system ideally suited for user authentication and structured, persistent data storage.



4.6 - Chatbot Features:

The NEXORA chatbot is designed to provide users with supportive, informative, and accessible mental health guidance. It uses Botpress as the chatbot platform and is configured with a mental health knowledge base based on **Abnormal Psychology** by Ronald J. Comer and Jonathan S. Comer. The chatbot focuses on psychoeducation, coping guidance, and empathetic support rather than diagnosis or replacement of mental health professionals.

4.6.1 Standard 1: Greeting

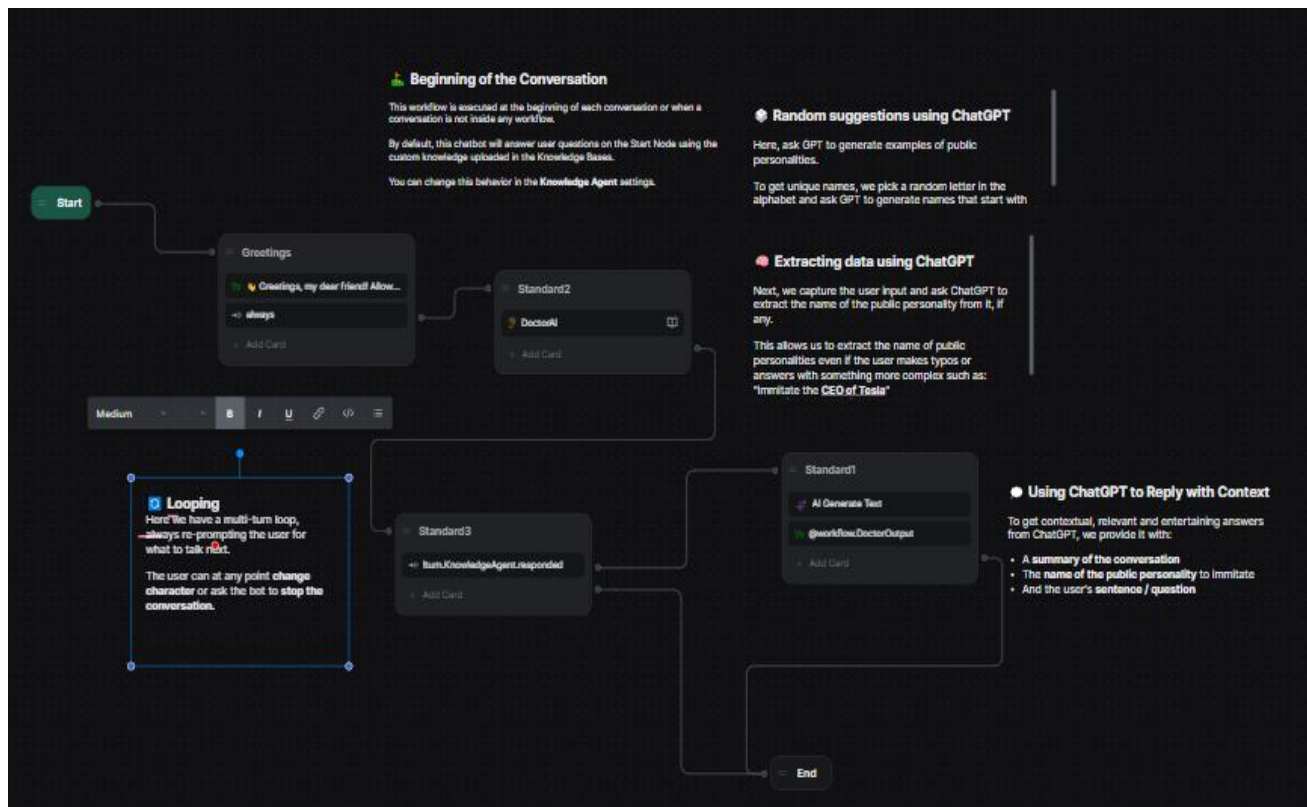
The chatbot begins the conversation with a friendly and welcoming greeting. This initial interaction creates a comfortable environment for users and encourages them to communicate openly about their concerns.

4.6.2 Standard 2: Response Using Knowledge Base

The chatbot uses **Botpress** to provide responses related to mental health and psychology. Its knowledge base is based on established psychological concepts and content from **Abnormal Psychology** by Ronald J. Comer and Jonathan S. Comer. It provides psychoeducational information, general coping strategies, and supportive responses while avoiding diagnosis or replacing professional psychological care.

4.6.3 Standard 3: Response to Out-of-Scope Queries

When a user's query falls outside the chatbot's configured mental health knowledge and support scope, the system can use its configured fallback mechanism to provide a suitable response. This helps maintain a useful conversation when users ask questions that are not directly covered by the chatbot's primary knowledge base.



4.7 - Implementation Phases:

4.7.1 Phase 1: Planning and Design

The initial phase focused on defining the purpose and scope of the NEXORA chatbot. The chatbot was planned as a supportive mental health guidance component rather than a diagnostic system. The required conversation flow, user interaction approach, and mental health topics to be covered were identified. A suitable knowledge source was also selected to provide reliable psychological information.

4.7.2 Phase 2: Development and Knowledge Base Integration

During this phase, the chatbot was developed and configured using **Botpress**. Conversation flows and user interactions were implemented according to the requirements of the NEXORA system. A mental health knowledge base based on *Abnormal Psychology* by Ronald J. Comer and Jonathan S. Comer was incorporated to support responses related to psychological concepts, psychoeducation, coping strategies, and general mental health guidance.

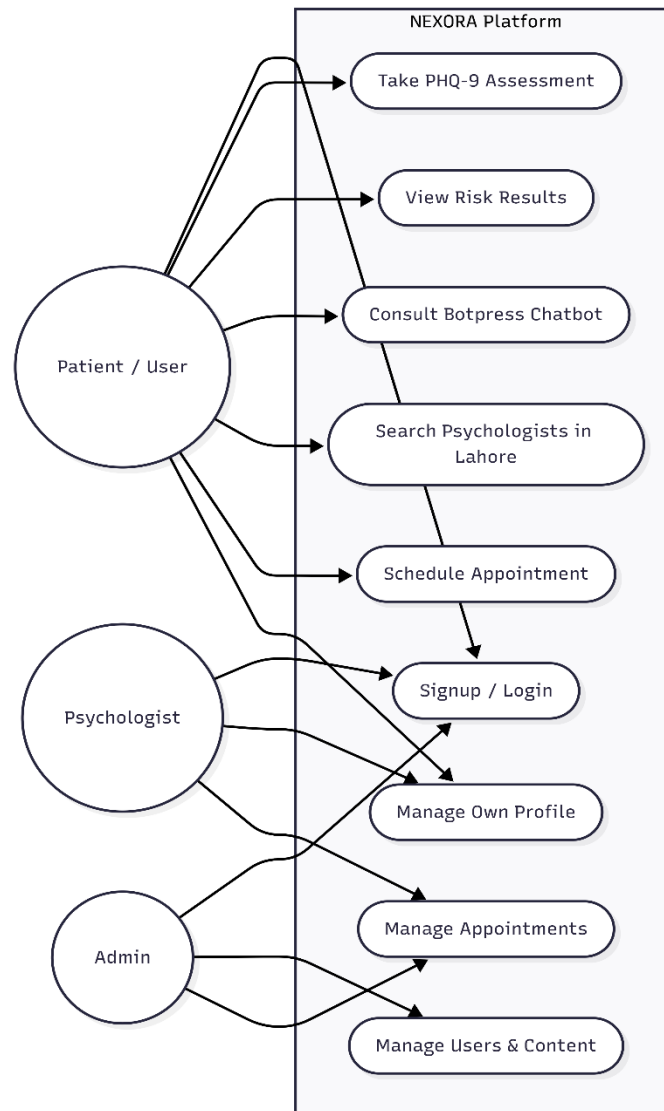
4.7.3 Phase 3: Testing and Validation

The chatbot was tested to verify that its conversation flow and responses worked as intended. Testing focused on common mental health queries, general user interaction, knowledge-based responses, greetings, and queries outside the chatbot's intended scope. The purpose of this phase was to ensure that the chatbot provided appropriate and supportive responses without presenting itself as a replacement for a mental health professional.

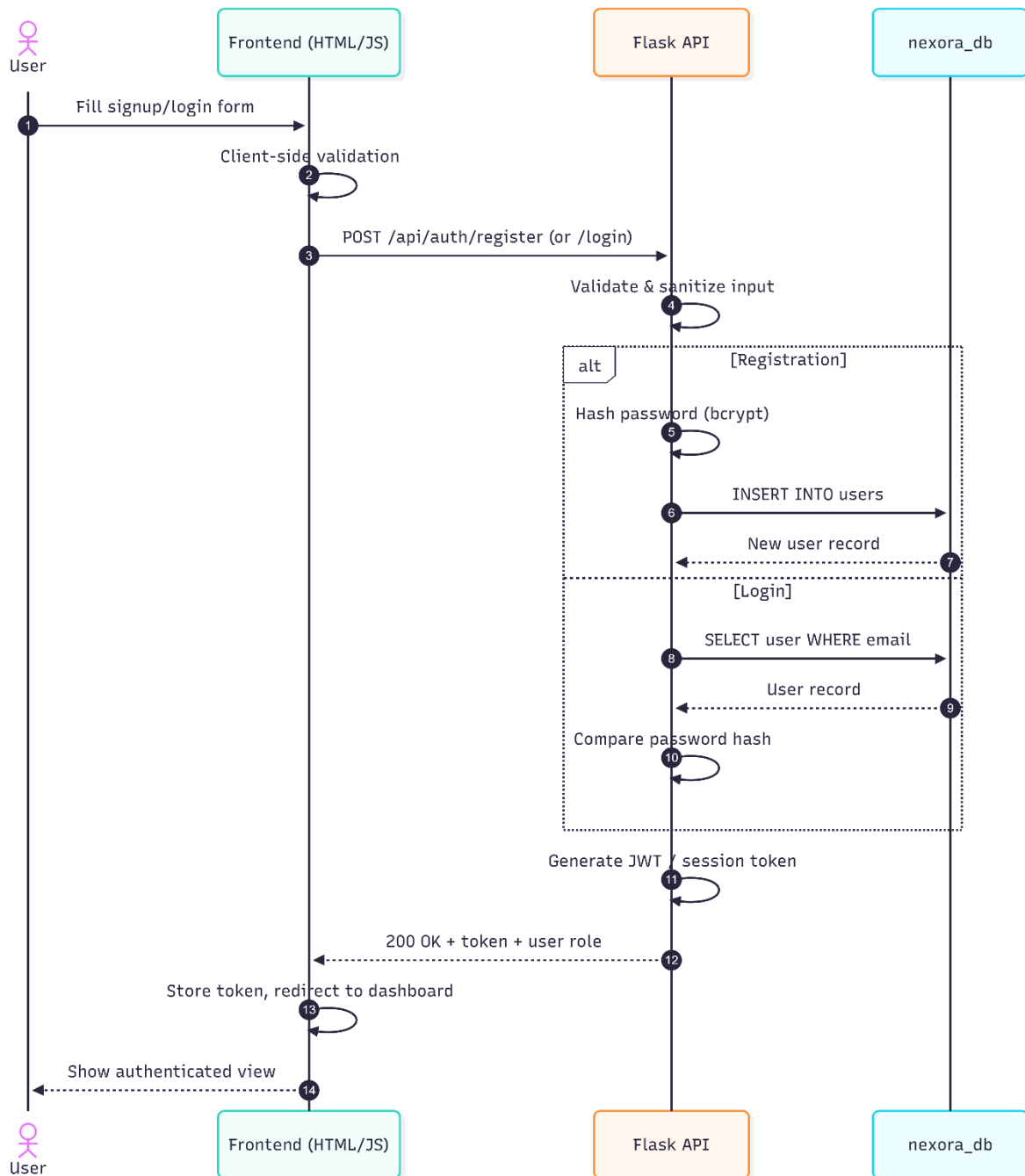
4.7.4 Phase 4: Deployment and Maintenance

In the final phase, the Botpress chatbot was integrated into the NEXORA website through the chatbot interface. The chatbot was made accessible through the dedicated chatbot page so that users could interact with it as part of the overall mental health guidance system. The chatbot configuration can be maintained and updated through Botpress when improvements or knowledge-base updates are required.

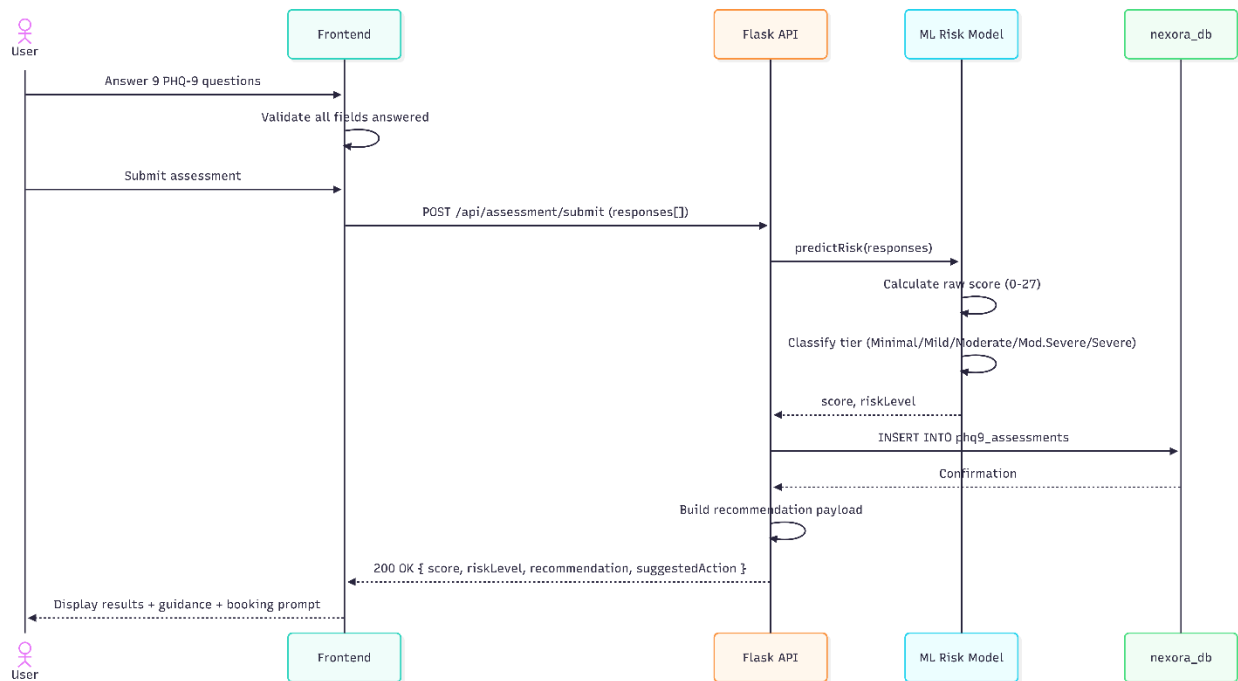
4.8 - System Use Case Diagram:



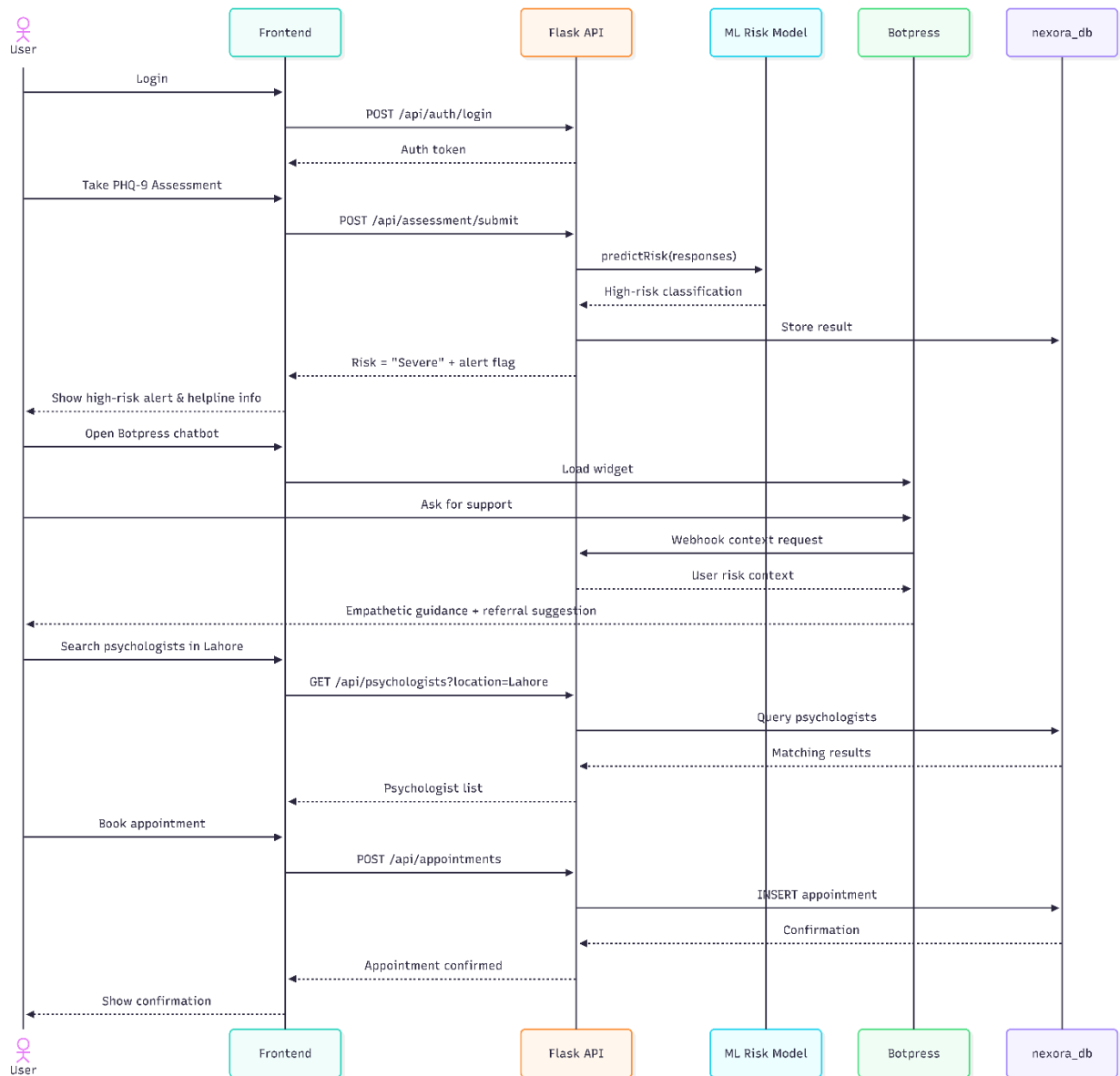
4.9 - Sign Up/ Sign In Sequence Diagram:



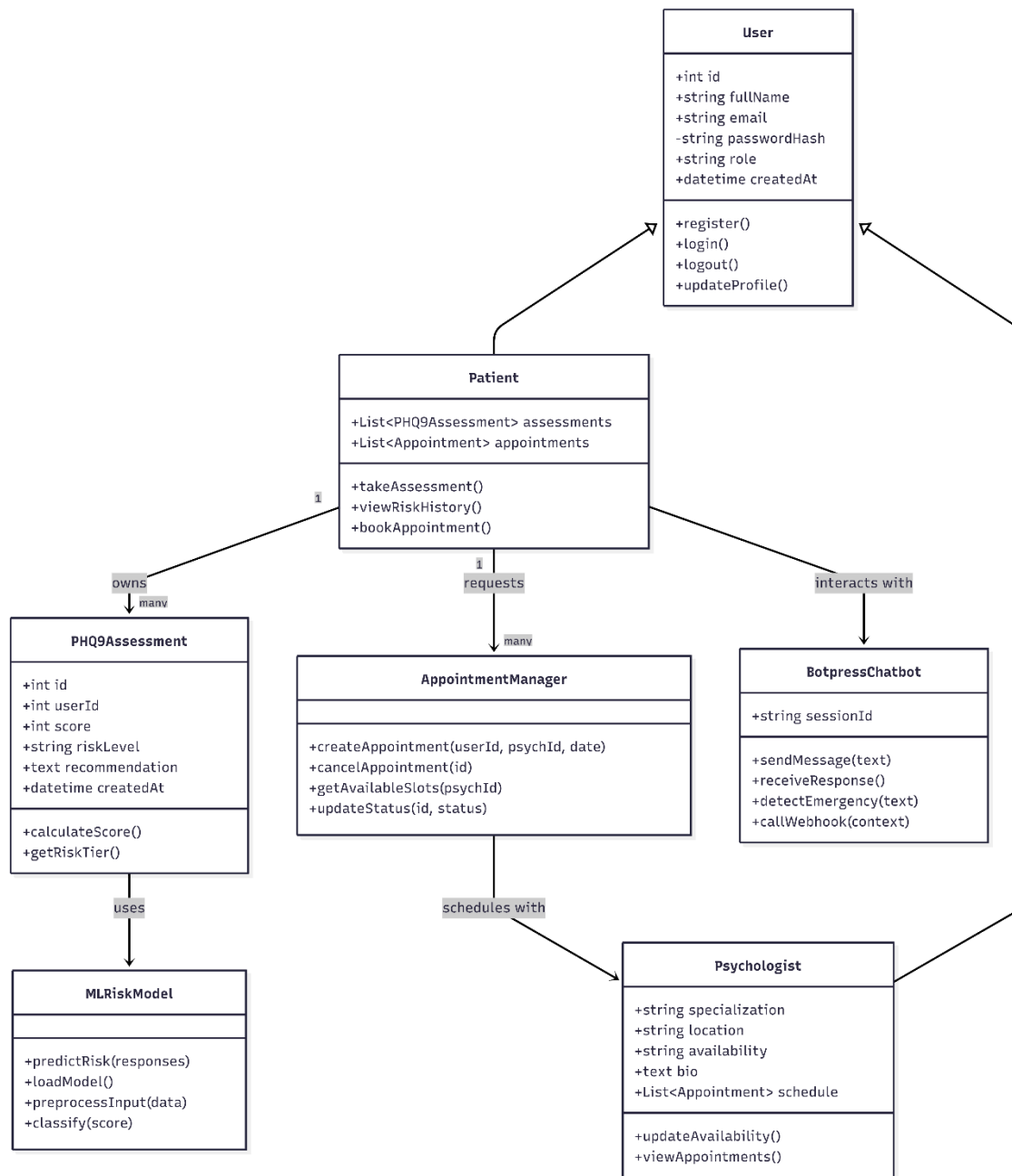
4.10 - PHQ-9 Assessment Sequence Diagram:



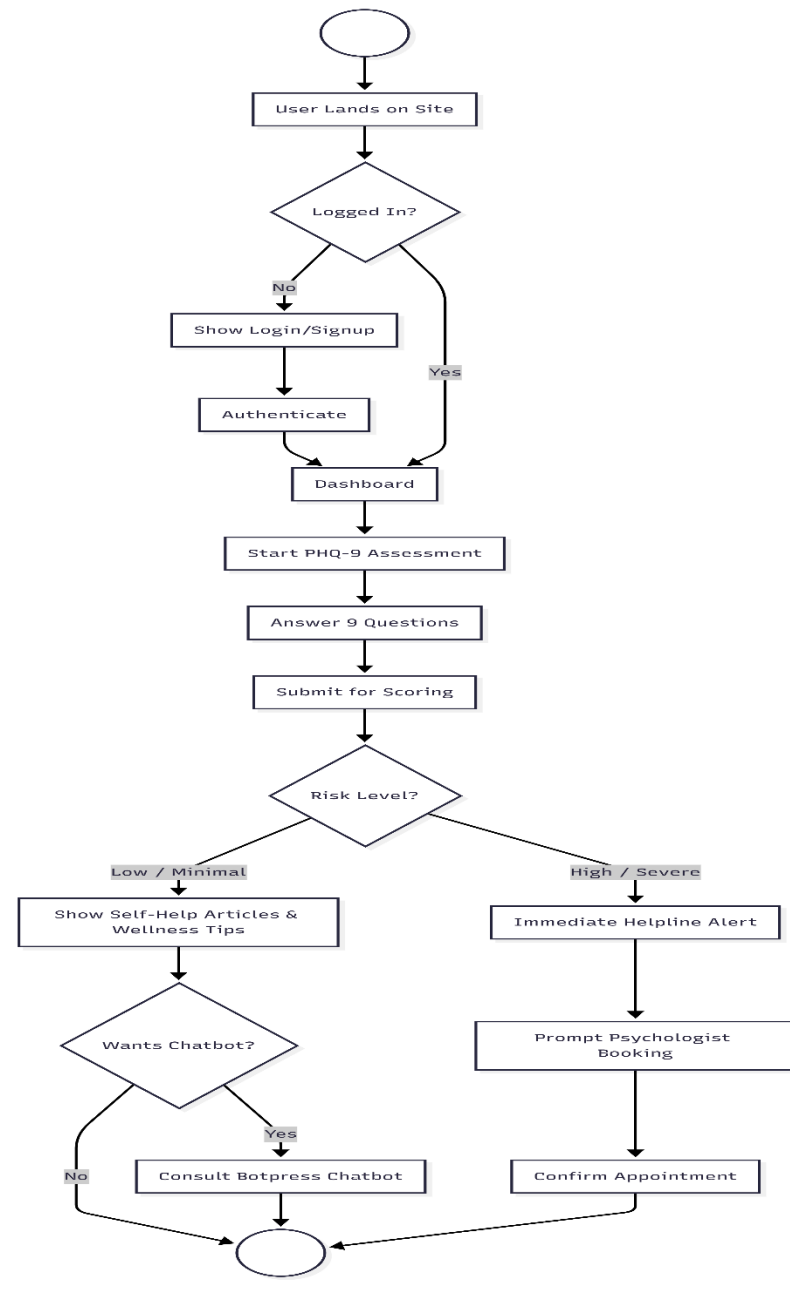
4.11 - General Sequence Diagram:



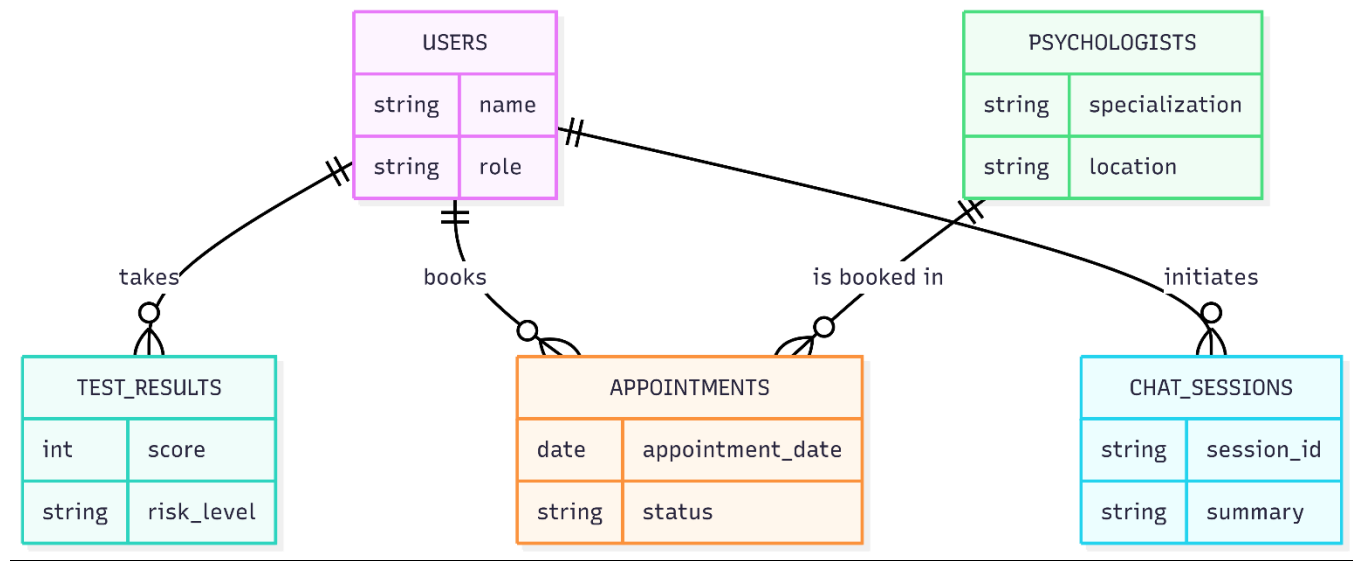
4.12 - Class Diagram:



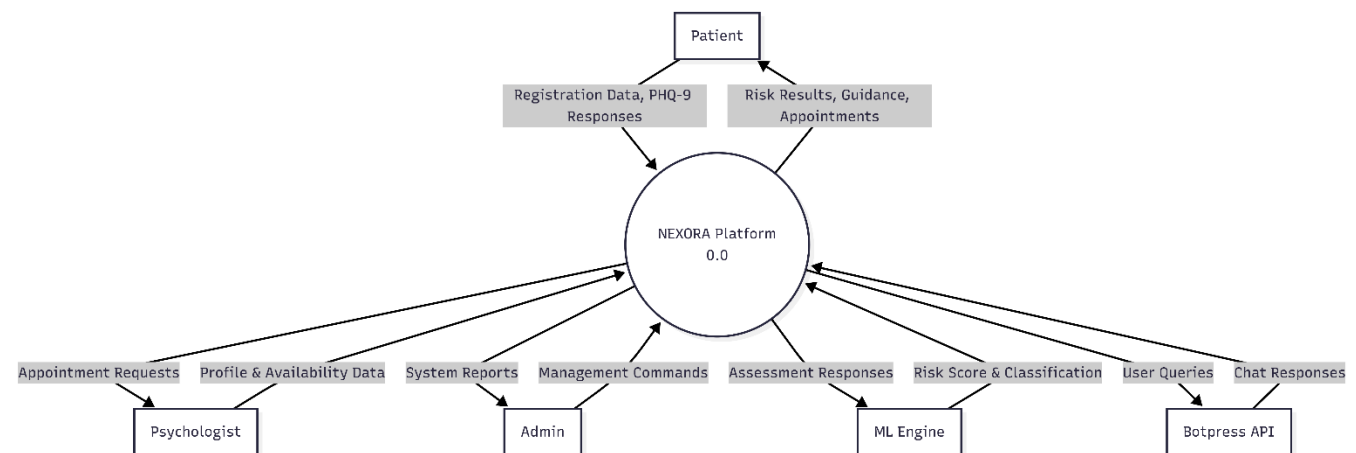
4.13 - Activity Diagram:



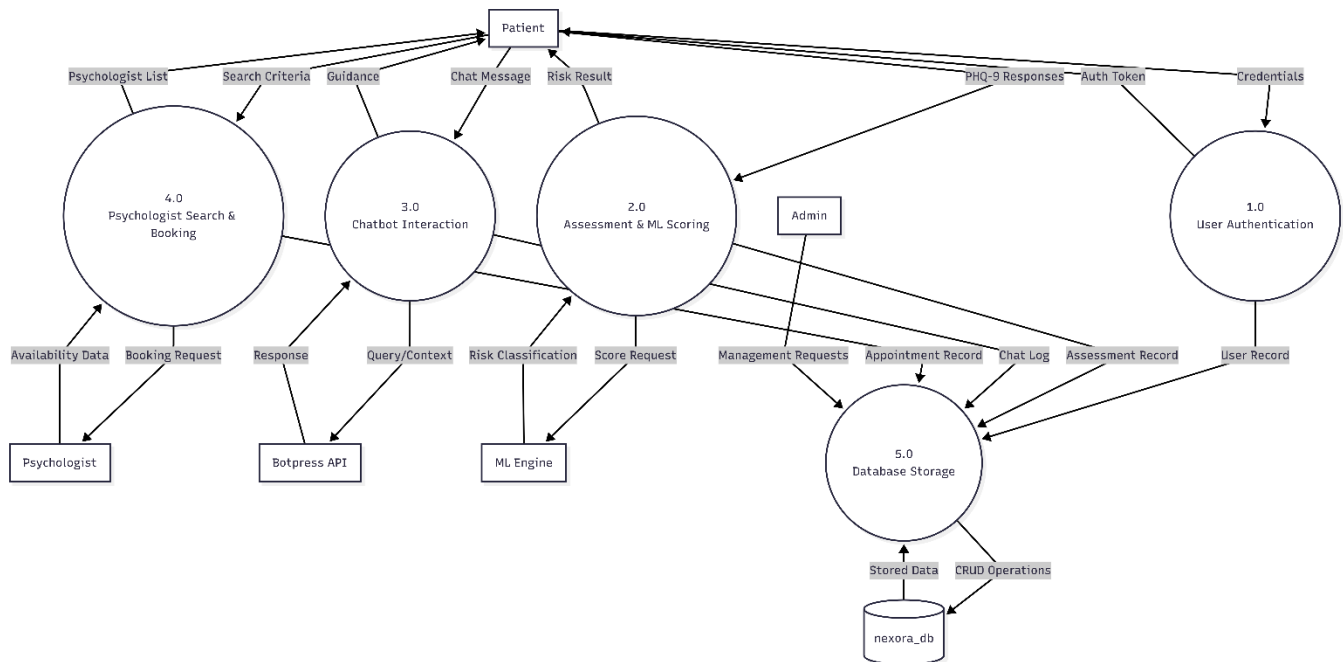
4.14 - Entity-Relationship Diagram:



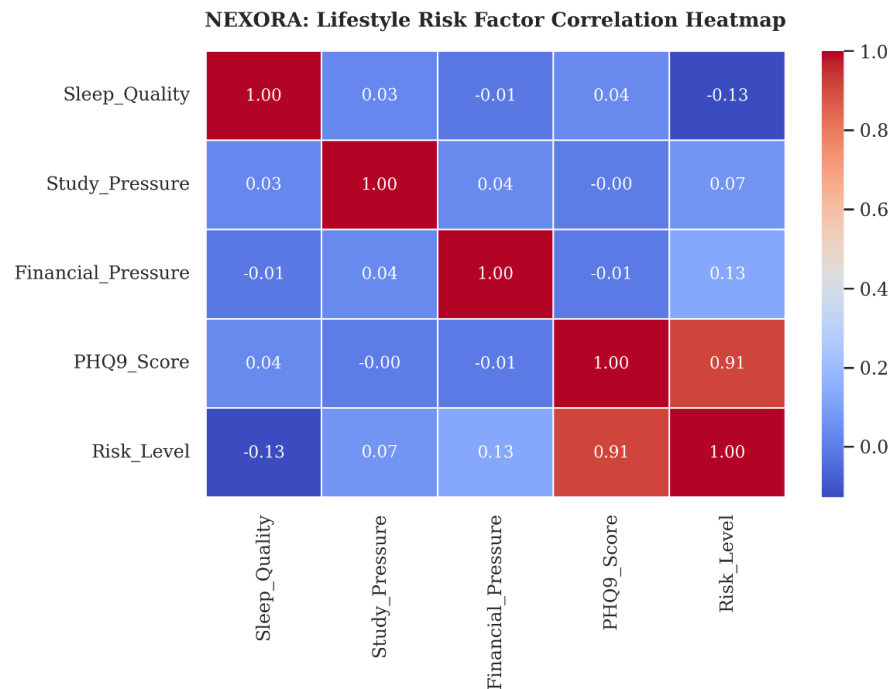
4.15 - Data Flow Diagram - DFD Level 0:



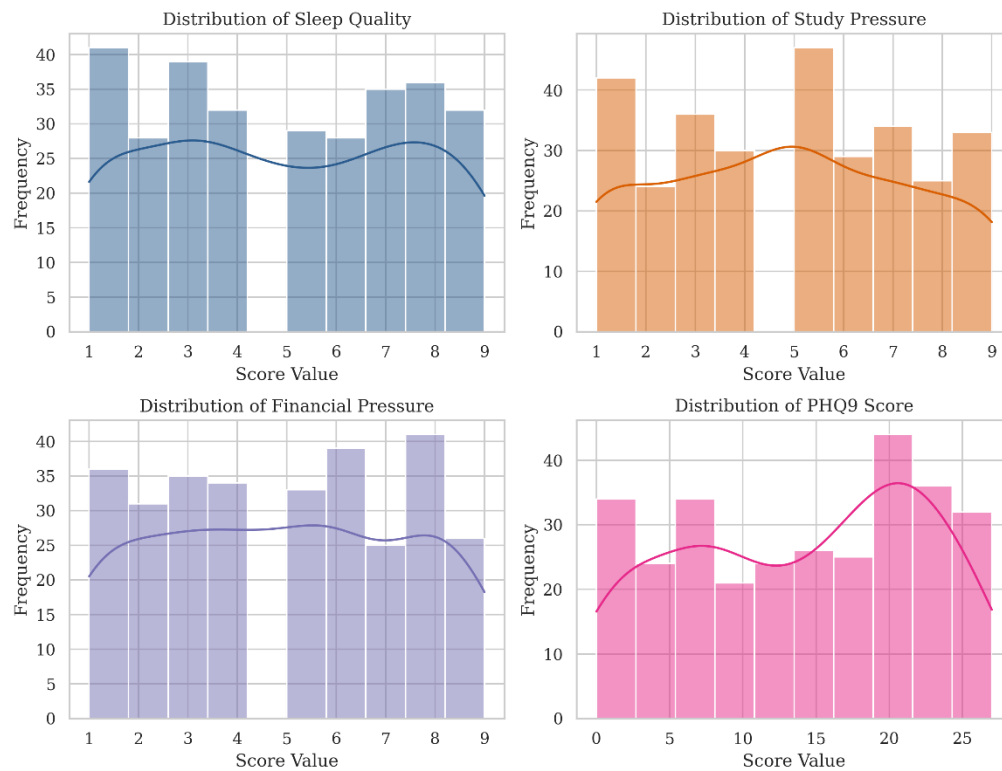
4.16 - Data Flow Diagram - DFD Level :



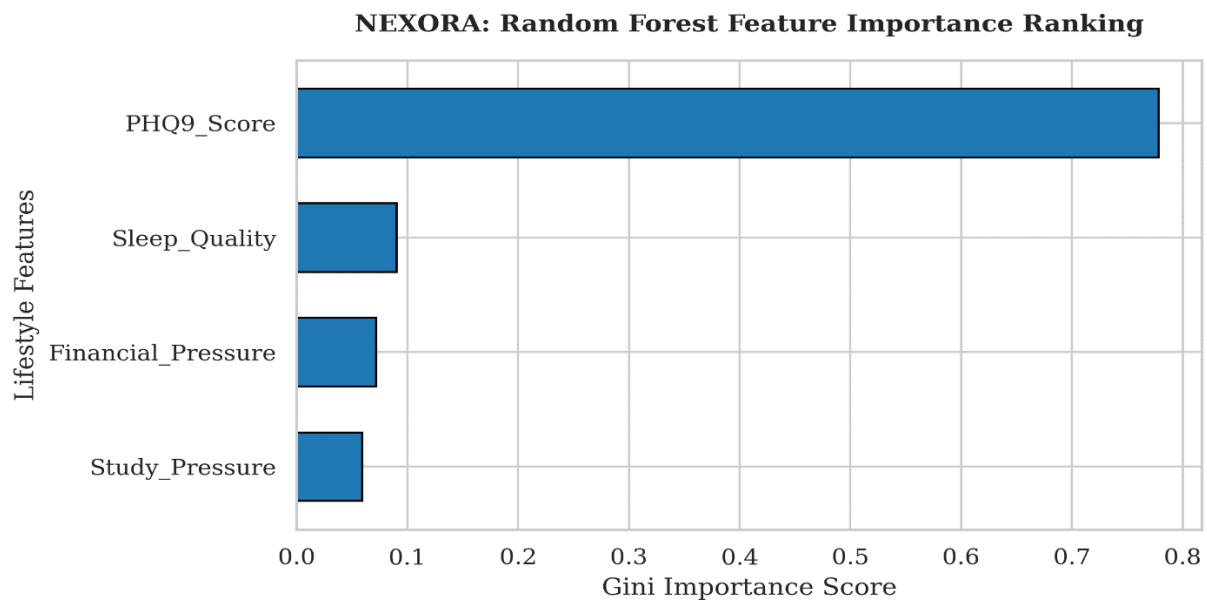
4.17 - Nexora Correlation Heatmap :



4.18 - Nexora Feature Histograms :



4.19 - Nexora Feature Importance:



4.20 - Conclusion:

The development of the NEXORA chatbot provides users with an accessible source of mental health information and supportive guidance. By integrating **Botpress** with a psychology-focused knowledge base derived from **Abnormal Psychology** by Ronald J. Comer and Jonathan S. Comer, the system can provide psychoeducation, general coping guidance, and empathetic support.

The chatbot is designed to **support users rather than diagnose mental health conditions or replace professional psychological care**. Its integration with the wider NEXORA system complements the PHQ-9 assessment, risk evaluation, and professional guidance features, providing users with a more complete mental health support experience.

Chapter No. 5
IMPLEMENTATION

5.1 - Overview:

This chapter provides a detailed description of how NEXORA was actually built. It covers the development environment, the Flask API backend, the PHQ-9 rule engine, the Random Forest lifestyle-risk model, the chatbot component, the frontend implementation, the database setup, and the security measures applied throughout.

The screenshot shows a Google Colab notebook titled 'Project_Final.ipynb'. The code in the first cell defines a Flask API with a single endpoint for GET requests that returns a JSON response indicating the API is online. The second cell runs the command 'pip install pyngrok' and shows the output indicating successful installation. The third cell imports ngrok from pyngrok and kills any previous tunnels.

```
[13] ✓ 0s
return jsonify({
    "status": "online",
    "message": "Mental Health Assessment API is running."
})

print("Flask API created successfully!")
print("Available endpoints:")
print("GET /")
print("POST /predict")

Flask API created successfully!
Available endpoints:
GET /
POST /predict

[14] ✓ 0s
!pip install pyngrok

Collecting pyngrok
  Downloading pyngrok-8.1.2-py3-none-any.whl.metadata (8.6 kB)
Requirement already satisfied: PyYAML>=5.1 in /usr/local/lib/python3.13/dist-packages (from pyngrok) (6.0.3)
  Downloading pyngrok-8.1.2-py3-none-any.whl (25 kB)
Installing collected packages: pyngrok
Successfully installed pyngrok-8.1.2

[15] ✓ 0s
from pyngrok import ngrok

# Close any previous tunnels
ngrok.kill()
```

The screenshot shows the continuation of the Google Colab notebook. The first cell prints 'Flask server started on port 5000'. The second cell configures ngrok with an authentication token and prints the configuration. The third cell creates a public tunnel to the Flask server on port 5000 and prints the public URL. The fourth cell imports requests and defines test data for the API endpoint.

```
Flask server started on port 5000

[17] ✓ 0s
from pyngrok import ngrok

ngrok.set_auth_token("3I3FCOH9s19zk2a7L0w2LHowStd_845J0cDTyLK7RtVjcwsv")
print("ngrok authentication configured.")

ngrok authentication configured.

[18] ✓ 0s
from pyngrok import ngrok

# Create a public tunnel to Flask
public_url = ngrok.connect(5000)

print("Your Flask API URL is:")
print(public_url)

... Your Flask API URL is:
NgrokTunnel: "https://eastcoast-bullfight-left.ngrok-free.dev" -> "http://localhost:5000"

[19] ✓ 0s
import requests

# Your public Flask API URL
API_URL = "https://eastcoast-bullfight-left.ngrok-free.dev/predict"

# Test assessment data
test_data = {
```

5.2 - Development Environment (Google Colab and ngrok):

The backend of NEXORA was developed and tested in Google Colab a free, cloud-based Python computing environment provided by Google. Colab was selected for this project for several practical reasons:

- It provides a ready-to-use Python environment without requiring complex local software installation.
- It allows Python scripts, Flask applications, and machine-learning models to be run directly in the browser.
- It is free of charge and does not require powerful local hardware.
- It supports Flask, Flask-CORS, scikit-learn, and joblib through simple pip installations.

Because a Flask server running inside Google Colab cannot be accessed directly from a regular browser or frontend JavaScript code (since it runs on Google's infrastructure, not a publicly accessible server), the team used ngrok during development and testing. ngrok is a tool that creates a secure tunnel from the public internet to a locally or Colab-hosted server. It generates a public HTTPS URL (for example, <https://abcd-1234.ngrok.io>) that forwards incoming requests to the Flask server inside Colab. This allowed the frontend JavaScript to send API requests to the Flask backend during development.

It is important to understand that this is strictly a development and testing arrangement. ngrok tunnel URLs change every time a session is started, and Colab sessions have a time limit. This arrangement is not suitable for a permanent production deployment. When ngrok is used, all API communication occurs over HTTPS, which encrypts data in transit.

5.3 - Backend Implementation, Python Flask:

Flask was chosen as the backend framework because it is a lightweight Python micro-framework that is easy to understand, quick to set up, and well-suited to building REST API endpoints for a project of this scope. Flask has excellent documentation, a large community, and strong support for the libraries used in this project. The fundamental communication pattern between the frontend and the Flask backend is:

User (Browser) → JavaScript Fetch → HTTP Request → Flask API → Processing → JSON Response
→ JavaScript Display

5.3.1 Key API Endpoints

POST /predict: The primary endpoint for the assessment feature. This endpoint is only reachable by an authenticated user; unauthenticated requests are rejected. The frontend collects all PHQ-9 answers and lifestyle selections, converts lifestyle selections to numbers (Good=0, Average=1, Bad=2, Worst=3), and sends the data as a JSON object. The endpoint: (1) receives and parses the JSON; (2) validates required fields; (3) calculates the PHQ-9 total score; (4) applies the rule engine for severity and self-harm flag; (5) passes lifestyle values to the Random Forest model; (6) collects the model's predicted classification and class probabilities; (7) writes the score, severity level, risk tier, and timestamp to the phq9_assessments table against the authenticated user_id; (8) returns a combined JSON response.

Health/Status Route: A basic route confirming the Flask server is running. Useful during development to verify the ngrok tunnel is correctly connected.

GET /history: Returns the authenticated user's past PHQ-9 assessment records from the phq9_assessments table, ordered by assessment_date. The frontend dashboard/history view uses this endpoint to render a user's assessment history. As with /predict, this endpoint requires an authenticated session and only ever returns records belonging to the requesting user.

5.3.2 Flask-CORS

Because the frontend and Flask backend may be served from different origins (especially during development with ngrok), Flask-CORS is used to enable Cross-Origin Resource Sharing. This allows the browser's JavaScript to make requests to the Flask API without being blocked by the browser's same-origin security policy.

5.4 - PHQ-9 Rule-Based Scoring Engine:

The Patient Health Questionnaire-9 (PHQ-9) is a clinically validated depression screening tool developed by Kroenke, Spitzer, and Williams (2001) and used worldwide by mental health professionals. The questionnaire consists of nine items, each asking about a common symptom of depression over the past two weeks. Response options range from "Not at all" (scored 0) to "Nearly every day" (scored 3). The total score ranges from 0 to 27.

Why the PHQ-9 was chosen: The decision to base the core assessment on the PHQ-9 was directly recommended by the medical professionals at Jinnah Hospital during the project's clinical validation phase (see Chapter 3). The PHQ-9 is clinically validated, globally and locally recognised, straightforwardly scored, and produces results that align with established clinical categories making it a far more responsible foundation for a mental health screening system than a purely AI-based assessment.

In NEXORA, the PHQ-9 total score is calculated by a rule-based engine in the Flask backend. The engine sums the nine item scores and applies the following standard severity bands:

Total Score Range	Severity Category
0 – 4	Minimal / No significant symptoms
5 – 9	Mild symptoms
10 – 14	Moderate symptoms
15 – 19	Moderately Severe symptoms
20 – 27	Severe symptoms

The rule engine additionally checks the response to PHQ-9 item 9 the item concerning thoughts of being better off dead or of hurting oneself. If the user's response to this item is anything other than "Not at all", the engine raises a self-harm/suicide-risk flag. This flag is included in the JSON response and displayed prominently on the results page with appropriate guidance urging immediate contact with a mental health professional or crisis helpline.

Important: NEXORA's PHQ-9 is presented as a screening tool, not a diagnostic instrument. A PHQ-9 score is not a clinical diagnosis. Users are clearly informed that results are an initial indication and that they should consult a qualified healthcare professional for any diagnosis or treatment.

5.5 - Supplementary Random Forest Lifestyle-Risk Model:

In addition to the PHQ-9, NEXORA includes a supplementary lifestyle-based risk component using a Random Forest machine-learning classifier. A Random Forest is an ensemble algorithm that constructs a large number of decision trees during training and combines their outputs to produce a final classification well suited to classification tasks with categorical or ordinal inputs like the lifestyle questions in NEXORA.

The three lifestyle inputs used by the model are:

- Sleep Quality: How well the user has been sleeping. Options: Good (0), Average (1), Bad (2), Worst (3).
- Study Pressure: Academic or work-related pressure level. Options: Good/Low (0), Average (1), Bad/High (2), Worst/Severe (3).
- Financial Pressure: Level of financial stress. Options: Good/Low (0), Average (1), Bad/High (2), Worst/Severe (3).

The frontend converts each option to its numeric value before sending the data to the Flask backend. The backend passes these three values to the trained Random Forest model (phq_risk_model.pkl), which returns a lifestyle-risk classification of Low, Moderate, or High along with class probabilities. This classification is displayed on the results page clearly labelled as a supplementary indicator.

Why these lifestyle factors: Sleep quality, academic pressure, and financial stress are well-documented contributors to mental health outcomes. By including them as supplementary inputs, the system provides a broader picture of the user's situation beyond what the PHQ-9 alone captures.

5.5.1 - The Distinction Between the Two Assessment Components

PHQ-9 (Rule Engine) Primary Component	Random Forest Lifestyle-Risk Supplementary Component
Standardised, clinically validated 9-item questionnaire. Scored by deterministic rule engine. Produces score (0–27) and severity category. Includes self-harm/suicide-risk flag. Recommended directly by medical professionals at Jinnah Hospital.	Machine-learning model using 3 lifestyle inputs. Produces supplementary risk classification (Low/Moderate/High). Not a clinical screening tool. Not equivalent to PHQ-9 result. Adds contextual depth only.

This design combining a structured clinical questionnaire with a supplementary AI-based lifestyle indicator rather than depending entirely on AI was the core insight from the professional guidance received at Jinnah Hospital. The PHQ-9 provides the structured, clinically grounded foundation; the lifestyle model adds contextual depth.

5.6 - Random Forest Model Training, Dataset and Performance

Evaluation:

5.6.1 Model Training Methodology

NEXORA uses a **Random Forest Classifier** as a supplementary machine-learning component for lifestyle-related mental health risk prediction. The Random Forest model does not calculate the PHQ-9 score or determine the official PHQ-9 severity category. Instead, the PHQ-9 score and severity are calculated independently through the rule-based PHQ-9 scoring engine, while the Random Forest model provides an additional lifestyle-risk prediction.

The model was trained using three input features:

- **Sleep Quality**
- **Study Pressure**
- **Financial Pressure**

These features were selected because they demonstrated stronger relationships with PHQ-9 severity compared with the demographic variables evaluated during feature analysis. PHQ-9 question responses and the PHQ-9 total score were not used as Random Forest input features because the PHQ-9 severity is directly derived from these responses. Excluding them prevents the machine-learning model from simply reproducing the same outcome.

The original five PHQ-9 severity categories were mapped into three broader lifestyle-risk categories:

Original PHQ-9 Severity	Lifestyle Risk Class
Minimal	Low
Mild	Low
Moderate	Moderate
Moderate Severe	High
Severe	High

The dataset was divided into **80% training data and 20% testing data** using a stratified split with `random_state = 42`. The resulting dataset contained **532 training records and 133 testing records**. The Random Forest model was configured with:

- **Algorithm:** Random Forest Classifier
- **Number of trees:** 100
- **Maximum tree depth:** 6
- **Random state:** 42
- **Training/testing split:** 80% / 20%
- **Cross-validation:** 5-fold
- **Input features:** 3
- **Output classes:** Low, Moderate, High

The trained model was saved as `phq_risk_model.pkl` using Joblib so that it could subsequently be loaded by the Flask backend during assessment processing.

5.6.2 Dataset Description and Preprocessing

The original dataset contained **668 records and 23 columns**. During preprocessing, **3 duplicate records were removed**, resulting in **665 records**. Six redundant or constant columns were subsequently removed during dataset preparation.

The final machine-learning training dataset used by the Random Forest implementation contains **665 records and 5 columns**, consisting of the three selected input features, the original PHQ-9 severity label, and its encoded form:

Column	Description
Sleep Quality	Ordinal lifestyle assessment value
Study Pressure	Ordinal study-related pressure value
Financial Pressure	Ordinal financial-pressure value
PHQ_Severity	Original PHQ-9 severity category
PHQ_Severity_Encoded	Numerical encoding of the severity category

The three lifestyle variables use an ordinal scale where the assessment options are encoded as:

- **0 = Good**
- **1 = Average**
- **2 = Bad**
- **3 = Worst**

For machine-learning purposes, the encoded PHQ-9 severity values were converted into three broader risk classes using the following mapping:

Column	Description
0 = Minimal	Low
1 = Mild	Low
2 = Moderate	Moderate
3 = Moderately Severe	High
4 = Severe	High

5.6.3 Feature Analysis

Feature analysis was performed before model training to identify variables that were more strongly associated with PHQ-9 severity.

Feature	Correlation	Mutual Information
Study Pressure	0.56	0.193
Sleep Quality	0.53	0.164
Financial Pressure	0.42	0.119
Age	0.03	0.026

Gender	-0.04	0.008
--------	-------	-------

Study Pressure demonstrated the strongest relationship among the evaluated variables, followed by Sleep Quality and Financial Pressure. These three variables were therefore used as the input features for the Random Forest model.

5.6.4 - Model Performance Evaluation:

The Random Forest model was evaluated using both **5-fold cross-validation** and a separate **20% test set**. **Cross-Validation Performance:** The model achieved a mean **5-fold cross-validation accuracy of 70.5%**, with a standard deviation of approximately **2.9 percentage points**:

Cross-validation accuracy = 0.705 ± 0.029 This indicates that the model achieved relatively consistent performance across the five validation folds. **Test-Set Performance:** On the separate 20% test set containing **133 records**, the model achieved an overall accuracy of **68.42%**.

The class-wise performance was:

Risk Class	Precision	Recall	F1-Score	Support
High	0.66	0.61	0.63	38
Low	0.73	0.92	0.81	71
Moderate	0.33	0.12	0.18	24
Overall Accuracy			0.68	133
Macro Average	0.57	0.55	0.54	133
Weighted Average	0.64	0.68	0.65	133

The results show that the model performed best for the **Low-risk** category, achieving a recall of **92%** and an F1-score of **81%**. The **High-risk** category achieved a recall of **61%** and an F1-score of **63%**. The Moderate-risk category was more difficult for the model to distinguish, with a recall of **12%** and an F1-score of **18%**.

The difference between the **70.5% cross-validation accuracy** and the **68.42% test-set accuracy** is expected because the two measurements are obtained using different evaluation procedures.

5.6.5 Confusion Matrix

The confusion matrix was generated using the same 80/20 stratified test split and the trained Random Forest model.

Actual \ Predicted	High	Low	Moderate
High	23	12	3
Low	3	65	3
Moderate	9	12	3

The matrix contains **133 test observations** in total. The model correctly classified **23 of 38 High-risk cases**, **65 of 71 Low-risk cases**, and **3 of 24 Moderate-risk cases**. The relatively large number of Moderate cases classified as Low or High explains the comparatively low recall for the Moderate class.

5.6.6 Random Forest Feature Importance

The trained Random Forest model provides feature-importance values indicating the relative contribution of each input feature to the model's decision-making process.

Feature	Feature Importance
Study Pressure	0.3805
Sleep Quality	0.3484
Financial Pressure	0.2711

The results show that **Study Pressure** had the highest feature importance at approximately **38.05%**, followed by **Sleep Quality** at approximately **34.84%** and **Financial Pressure** at approximately **27.11%**. These values represent the Random Forest's calculated feature importance and should not be confused with the correlation values presented during feature analysis. Both measurements provide different types of information about the relationship between the variables and the model's predictions.

5.6.7 Model Integration with NEXORA

After training and evaluation, the Random Forest model was saved as `phq_risk_model.pkl` and integrated into the Flask backend. During an assessment, the frontend sends the user's Sleep Quality, Study Pressure, and Financial Pressure values to the Flask `/predict` endpoint. The Flask backend processes the PHQ-9 responses through the **rule-based PHQ-9 scoring engine** and processes the three lifestyle variables through the **Random Forest model**. The resulting response contains the PHQ-9 score, PHQ-9 severity, suicide-risk flag, lifestyle-risk category, and lifestyle-risk probabilities.

This separation allows NEXORA to use the validated PHQ-9 scoring rules as the authoritative screening result while using machine learning as a **supplementary lifestyle-risk assessment component**.

5.7 - Chatbot Component:

The NEXORA chatbot is a supportive, conversational component designed to provide users with psychoeducation, general mental-health information, and evidence-based coping strategies. It is not a clinical tool and must not be understood as replacing a psychiatrist, psychologist, doctor, therapist, or emergency service.

5.7.1 Platform Botpress

The chatbot is built using the Botpress platform an open-source conversational AI framework that supports the creation of chatbots with custom knowledge bases. Botpress allows the chatbot to be trained on domain-specific content, enabling it to answer questions about mental health topics in a way grounded in established psychological literature rather than relying on a general-purpose AI model.

5.7.2 Knowledge Base

The chatbot's primary source of information is a prepared knowledge base derived from *Abnormal Psychology* by Ronald J. Comer and Jonathan S. Comer a comprehensive academic text in abnormal psychology. Relevant chapters covering mood disorders, depressive disorders, symptoms of depression, psychoeducation, coping strategies, and supportive guidance were used to build the knowledge base. This allows the chatbot to respond to questions about what depression is, how it manifests, coping strategies, the importance of sleep and social connection, when to seek professional help, and what to expect from mental health treatment.

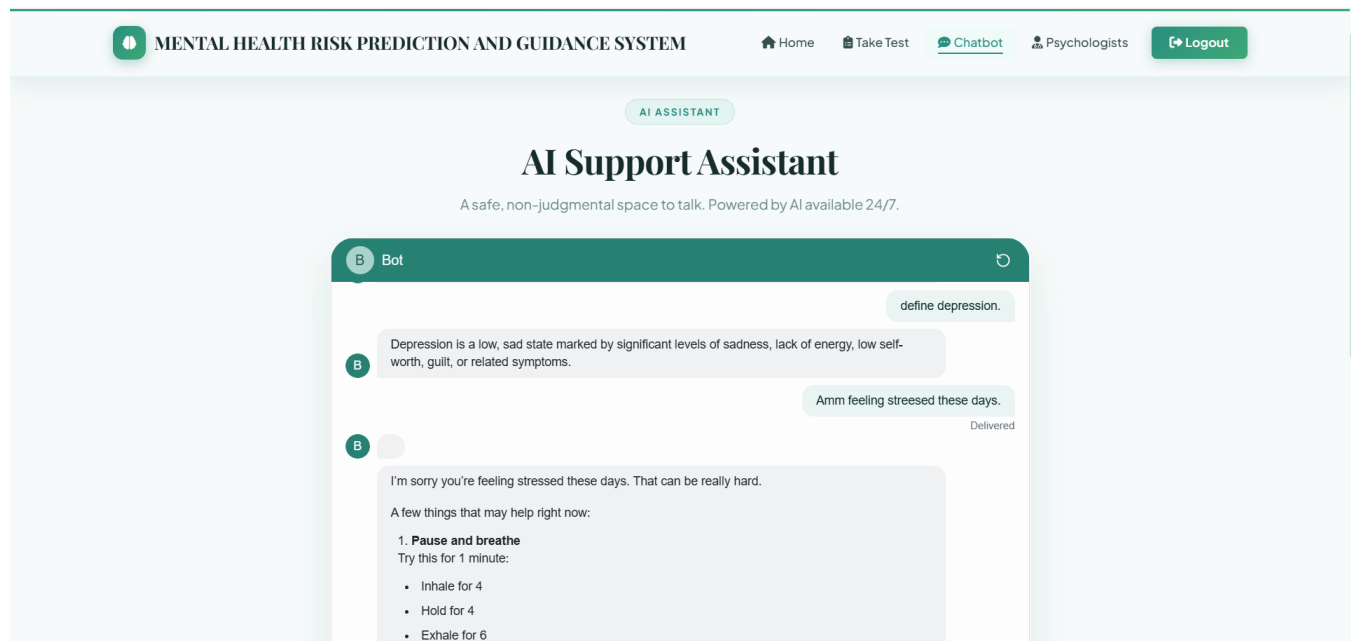
5.7.3 Interaction Flow and Fallback

When a user sends a message, the system first searches the knowledge base for a relevant response. If a relevant match is found with sufficient confidence, the chatbot responds from the knowledge base. If the query falls outside the scope of the knowledge base (for example, if it is unrelated to mental health), the system uses an external AI service as a fallback to handle the query gracefully rather than ignoring it.

Important: Relevant mental health queries → Knowledge Base → response. Irrelevant or out-of-scope queries → Fallback AI service → response.

5.7.4 Safety and Responsible Design

- The chatbot does not diagnose any mental health condition. All responses are educational and supportive.
- For queries indicating distress or risk, the chatbot encourages the user to contact a mental health professional or, in cases of immediate risk, a crisis helpline or emergency service.
- The chatbot never presents its responses as equivalent to a professional clinical assessment.
- Crisis helpline information relevant to Pakistan is included where appropriate.



5.8 - Frontend Implementation:

The frontend of NEXORA is implemented using HTML5, CSS3, JavaScript (ES6), and Bootstrap 5.

5.8.1 Bootstrap 5

Bootstrap 5 is a popular open-source CSS framework providing a responsive grid system, pre-built UI components (cards, buttons, forms, navigation bars), and utility classes. Using Bootstrap 5 reduces the amount of custom CSS required and ensures the layout automatically adapts to different screen sizes.

5.8.2 Pages

- **index.html** (Home Page): Introduces the system, explains what NEXORA does, and directs users to take the assessment or log in.
- **assessment.html** (Assessment Page): Accessible only to logged-in users; an unauthenticated visitor is redirected to auth.html. Presents the PHQ-9 questionnaire (9 items) and lifestyle questions (3 items). Collects and submits data to /predict via a Fetch API call.
- **Results Section**: Dynamically displays PHQ-9 score, severity label, self-harm flag (if applicable), supplementary lifestyle-risk classification, guidance text, and disclaimer. Clearly distinguishes PHQ-9 from lifestyle-risk results.
- **chatbot.html** (Chatbot Page): Hosts the Botpress chatbot interface for supportive conversation.
- **psychologists.html** (Professional Guidance Page): Displays information about mental health professionals.
- **auth.html** (Login / Registration Page): Contains both forms switched via tabs. Handles account creation and login.
- **(Booking Page)**: Allows users to explore appointment booking.
- **history.html** (Assessment History / Dashboard Page): Accessible only to logged-in users. Displays the user's past PHQ-9 assessments retrieved from GET /history, including score, severity label, risk tier, and date for each past submission.
- **crisis-helpline.html** (Crisis Helpline Page): Displays local 24/7 emergency mental health contacts, including Rescue 1122, Edhi Foundation, Umang Helpline, Rozan Counseling, and the National Youth Helpline. Accessible via main navigation and the footer.
- **articles.html** (Self-Help Articles Page): Presents categorised wellness resources covering Anxiety, Depression, Mindfulness, Sleep, and Coping Strategies.
- **terms-of-use.html** (Terms of Use Page): Outlines medical disclaimers, eligibility, liability limitations, and acceptable use of the platform.
- **privacy-policy.html** (Privacy Policy Page): Details data collection practices, secure password storage using bcrypt hashing, use of the Google Colab/ngrok research server during development, and confidentiality policies.

5.8.3 JavaScript Frontend-Backend Communication

JavaScript using the browser's native Fetch API handles all communication with the Flask backend. When the user submits the assessment form, JavaScript collects form values, encodes lifestyle selections numerically, builds a JSON object, and sends it asynchronously to /predict using fetch(). When the

response is received, JavaScript parses the JSON and dynamically updates the results section without a full page reload. The same pattern applies for authentication: form inputs are collected and sent to the appropriate Flask endpoint, with the response handled to redirect the user or display an error message.

5.9 - Database Implementation:

The MySQL database for NEXORA is named `nexora_db`. It stores user registration data in the `users` table with passwords stored as secure hashes using `bcrypt`. Every completed PHQ-9 assessment is additionally written to the `phq9_assessments` table, linked to the submitting user via `user_id`, so that results persist across sessions and can be retrieved for the history/dashboard view. The Flask backend connects to the MySQL instance using a Python MySQL connector library. All connection parameters (host, user, password, database name) are stored as environment variables. All database queries use parameterised statements (prepared statements) to prevent SQL injection vulnerabilities.

5.10 - Security Considerations:

- Password Hashing: User passwords are never stored as plain text. A cryptographic hashing function (`bcrypt`) transforms each password before storage, so even if the database were accessed unauthorised, actual passwords cannot be recovered.
- Environment Variables: Database connection parameters and any API keys are stored as environment variables, not hardcoded in source code.
- Parameterised SQL Queries: All database queries use parameterised statements to prevent SQL injection attacks.
- UNIQUE Email Enforcement: The email column has a UNIQUE constraint preventing duplicate registrations.
- HTTPS via ngrok: During development, all API communication occurs over HTTPS through ngrok's secure tunnel.
- Session-Gated Assessment Access: The PHQ-9 assessment and assessment history endpoints require an authenticated session; anonymous requests are rejected, and each user can only retrieve their own records from the `phq9_assessments` table.

Chapter No. 6

TESTING

6.1 - Overview:

Testing is a critical quality assurance phase that validates the functional correctness, integration integrity, and robustness of the system. This chapter documents the test cases for NEXORA, covering authentication, the PHQ-9 rule engine, the Random Forest endpoint, the chatbot, the frontend, and error handling. Where results are confirmed they are recorded; where results are pending, placeholders are provided.

6.2 - Unit Testing:

6.2.1 Authentication Module

TC ID	Description	Input	Expected Output	Actual	Status
UT01	Register with valid unique credentials	Name, new email, valid password	User record created; password stored as hash	[CONFIRM]	[CONFIRM]
UT02	Register with duplicate email	Already-registered email	Registration rejected with error	[CONFIRM]	[CONFIRM]
UT03	Login with correct credentials	Valid email + correct password	Authentication successful	[CONFIRM]	[CONFIRM]
UT04	Login with wrong password	Valid email + wrong password	Authentication rejected	[CONFIRM]	[CONFIRM]

6.2.2 PHQ-9 Rule Engine

TC ID	Description	Total Score Input	Expected Severity	Actual	Status
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UT05	Low PHQ-9 total (Minimal)	3	Minimal; no flag	[CONFIRM]	[CONFIRM]
UT06	Mild PHQ-9 total	7	Mild; no flag	[CONFIRM]	[CONFIRM]
UT07	Moderate PHQ-9 total	12	Moderate	[CONFIRM]	[CONFIRM]
UT08	Moderately Severe PHQ-9	17	Moderately Severe	[CONFIRM]	[CONFIRM]
UT09	Severe PHQ-9 total	24	Severe; urgent guidance	[CONFIRM]	[CONFIRM]
UT10	Item 9 > 0 (self-harm flag)	Item 9 = 1 or higher	Self-harm flag raised	[CONFIRM]	[CONFIRM]

6.2.3 Random Forest Lifestyle-Risk Endpoint

TC ID	Description	Lifestyle Inputs	Expected Classification	Actual	Status
UT11	All-good lifestyle	sleep=0, study=0, fin=0	Low risk	[CONFIRM]	[CONFIRM]
UT12	All-worst lifestyle	sleep=3, study=3, fin=3	High risk	[CONFIRM]	[CONFIRM]
UT13	Mixed lifestyle values	sleep=1, study=2, fin=1	Moderate risk	[CONFIRM]	[CONFIRM]

6.3 - Integration Testing:

TC ID	Workflow	Steps	Expected Outcome	Status
IT01	Full Assessment	Open assessment → answer all 12 questions → submit	PHQ-9 score + severity + lifestyle-risk classification displayed correctly	[CONFIRM]
IT02	Self-Harm Flag	Submit PHQ-9 with item 9 > 0	Self-harm flag displayed prominently with crisis guidance	[CONFIRM]
IT03	Register then Login	Register → logout → login with same credentials	Both registration and login succeed	[CONFIRM]

IT04	Chatbot Knowledge Base	Send mental-health question → send off-topic question	KB response for mental health; fallback for off-topic	[CONFIRM]
IT05	Disclaimer Display	Complete assessment → view results	PHQ-9 and lifestyle-risk clearly distinguished; disclaimer visible	[CONFIRM]

6.4 Frontend Testing:

TC ID	Test	Expected	Status
FT01	Pages load on Chrome, Firefox, Edge	All pages render correctly	[CONFIRM]
FT02	Responsive layout at 375px (mobile)	Layout adapts; all elements readable	[CONFIRM]
FT03	Submit assessment with unanswered questions	Validation prevents submission; user prompted	[CONFIRM]
FT04	Navigation between pages	All links work correctly	[CONFIRM]

6.5 Error Handling and Edge Cases:

TC ID	Scenario	Expected Behaviour	Status
EH01	Flask backend unavailable (ngrok expired)	Frontend displays error message; does not crash silently	[CONFIRM]
EH02	Submit assessment with missing answers	Frontend validation prevents submission; user prompted to complete	[CONFIRM]
EH03	Register with already-registered email	Backend returns 409 or error; frontend displays appropriate message	[CONFIRM]
EH04	Submit invalid data types to /predict	Backend returns validation error; frontend handles gracefully	[CONFIRM]

Chapter No. 7
SYSTEM DEPLOYMENT

7.1 - Deployment Strategy:

After completing the development, integration, and testing of the NEXORA system, the project was prepared for deployment. The Flask backend was configured to provide the assessment API, while the frontend, database, chatbot, and machine-learning components were integrated into the complete system. The system was tested to ensure that the major functionalities, including user authentication, PHQ-9 assessment, supplementary lifestyle-based risk prediction, chatbot interaction, and appointment booking, worked correctly.

During development and testing, **Google Colab** and **ngrok** were used to run and expose the Flask-based machine-learning API for communication with the frontend. After completing the development and verifying the system, the project's source code and relevant project files were **uploaded to GitHub** for version control, project management, backup, and future maintenance. GitHub also provides a central repository from which the development team can maintain and update the project as required.

7.1.1 Current Development and Testing Deployment

During development and testing, NEXORA's Flask backend ran inside Google Colab and was made accessible to the frontend using ngrok. This arrangement was suitable for development, testing, and demonstration, but is not a permanent production deployment solution.

- Flask backend runs inside a Google Colab notebook.
- ngrok generates a public HTTPS URL tunnelling requests to the Colab-hosted Flask server.
- MySQL database runs on a cloud provider with credentials in environment variables.
- Frontend files are served locally or via a simple HTTP server.
- The frontend's API base URL must be updated each time the Colab session and ngrok tunnel are restarted.
- The project's source code is uploaded to GitHub for version control, source-code backup, maintenance, and future development. GitHub is used as a code repository and should not be interpreted as the production hosting environment.

7.1.2 Recommended Production Deployment

- **Flask Backend:** Deploy the Flask application to a reliable cloud hosting platform that provides persistent server execution and HTTPS support.
- **Machine Learning Model:** The `phq_risk_model.pkl` file should be included with the backend deployment, together with compatible versions of Python, scikit-learn, joblib, and other required dependencies.
- **MySQL Database:** The MySQL database should be hosted using a reliable managed database service. Database credentials should be stored securely using environment variables rather than being written directly into the source code.

- **Frontend:** The HTML, CSS, JavaScript, and Bootstrap files can be served through a static hosting service or through the Flask application, depending on the final deployment architecture.
- **Botpress Chatbot:** The Botpress chatbot should remain connected to the deployed frontend through its configured embedded-chat integration.
- **API URL:** The frontend should use the permanent HTTPS address of the deployed Flask backend instead of the temporary ngrok URL used during development.
- **GitHub Repository:** The final project source code should remain maintained in GitHub to support version control, backup, future modifications, and collaborative maintenance. GitHub serves as the project's source-code repository rather than automatically being considered the application's production server.
- **Security Configuration:** Production credentials, database connection details, and API keys should be stored through secure environment variables or the hosting platform's secret-management facilities.

7.1.3 Conversion Strategy

Since NEXORA is a newly developed system and does not replace an existing legacy application, a **Direct Cutover** strategy is appropriate for the initial production deployment. There is no existing production system that needs to operate alongside NEXORA or from which historical application data must be migrated.

The conversion process can therefore follow these stages:

1. **Prepare the Production Environment:** Set up the permanent Flask hosting environment, MySQL database, required dependencies, and necessary environment variables.
2. **Deploy the Backend and Model:** Upload the Flask application and the trained `phq_risk_model.pkl` model to the production environment and verify that the `/predict` endpoint operates correctly.
3. **Configure the Frontend:** Replace the development ngrok API address with the permanent production API URL.
4. **Connect Supporting Components:** Verify the Botpress chatbot, MySQL authentication, assessment workflow, and professional-guidance/booking pages.
5. **Perform Final Testing:** Test registration, authentication, PHQ-9 scoring, lifestyle-risk prediction, results display, chatbot interaction, and other implemented functions before public release.

6. **Initial Release:** Deploy NEXORA for its intended users after successful verification. A limited initial release, such as to the college community, can be used to identify usability or operational issues before wider availability.
7. **Maintain the Source Code:** Keep the final implementation and subsequent updates in the project's GitHub repository so that changes can be tracked and the source code remains available for future maintenance.

7.2 Anticipated Challenges and Solutions:

Challenge	Description	Solution
Session/Tunnel Stability	Google Colab sessions are temporary, and ngrok tunnel URLs can change when a session is restarted, which can interrupt communication between the frontend and Flask backend.	Migrate the Flask backend to a stable, persistent hosting environment before production deployment.
Model Portability (phq_risk_model.pkl)	Serialized for a specific Python/scikit-learn version; version mismatch causes loading errors.	Pin exact Python, scikit-learn, and joblib versions in requirements.txt.
Secrets Management	Sensitive credentials or API keys used during development could be exposed if they are stored directly in notebook cells or source files.	Move all secrets to environment variables before any production deployment.
CORS Configuration	The current Flask application uses CORS to allow communication between the frontend and backend during development. Production deployment may require more restrictive origin settings.	Restrict CORS to the approved production frontend domain when the system is deployed publicly.

Chapter No. 8
RESULTS

8.1 - System Features Delivered:

8.1.1 User Registration and Authentication

The system allows new users to create an account with full name, email, and password. The password is stored as a secure hash in the MySQL nexora_db database. Returning users can log in with email and password verified against the stored hash. The UNIQUE constraint on the email field prevents duplicate registrations.

The interface for the MENTAL HEALTH RISK PREDICTION AND GUIDANCE SYSTEM (NEXORA) features a navigation bar with links to Home, Take Test, Chatbot, Psychologists, and a Login button. The main content area is divided into two sections: a welcome message and a sign-in/registration form.

Welcome to NEXORA
Your journey toward better mental wellbeing begins here. Access personalised assessments, AI support, and professional care.

- 100% private & confidential
- Track your progress over time
- 24/7 AI support assistant
- Connect with psychologists

Sign in to your account
Or create a free account below

Log In Create Account

Full Name
Your full name

Email Address
you@example.com

Password
At least 8 characters

Confirm Password
Repeat your password

By creating an account you agree to our Terms of Service and Privacy Policy.

Create Free Account

The bottom screenshot shows the same interface but with the 'Log In' button highlighted and the 'Create Account' button disabled, indicating the user is logging in rather than registering.

8.1.2 PHQ-9 Structured Mental Health Assessment

The system presents users with the standard nine-item PHQ-9 questionnaire through a clean interface. Each question has four response options (Not at all=0; Several days=1; More than half the days=2; Nearly every day=3). The Flask backend calculates the total score and classifies it into one of five severity categories via the rule engine. If item 9 indicates any level of self-harm or suicidal ideation, the self-harm/suicide-risk flag is displayed prominently with crisis guidance.

8.1.3 Supplementary Lifestyle-Risk Assessment

Three lifestyle questions (Sleep Quality, Study Pressure, Financial Pressure) are presented after the PHQ-9. Selections are encoded numerically and passed to the Random Forest model. The model returns a lifestyle-risk classification (Low, Moderate, or High) displayed on the results page as a supplementary indicator alongside the PHQ-9 result.

A How would you describe your sleep quality?

Good Average Bad Worst

B How would you describe your current study pressure?

Good Average Bad Worst

C How would you describe your current financial pressure?

Good Average Bad Worst

Important: This screening tool is not a diagnostic instrument. If you are experiencing thoughts of self-harm, please contact a crisis helpline or emergency services immediately.

[Check My Mental Health](#)

8.1.4 Combined Results Display

The results page shows the PHQ-9 total score, severity label, self-harm/suicide-risk flag status, and supplementary lifestyle-risk classification. Both results are clearly labelled and distinguished. Appropriate guidance accompanies each result, encouraging professional consultation for elevated scores.

MENTAL HEALTH RISK PREDICTION AND GUIDANCE SYSTEM

Home Take Test Chatbot Psychologists Login

Severe Depressive Symptoms

Your PHQ-9 responses indicate severe depressive symptoms. Please seek professional mental health support promptly. Your response to the self-harm question indicates that additional support should be sought promptly.

27

PHQ-9 Score — Severe

Score range: 0-27 | Scores ≥ 10 suggest clinical significance

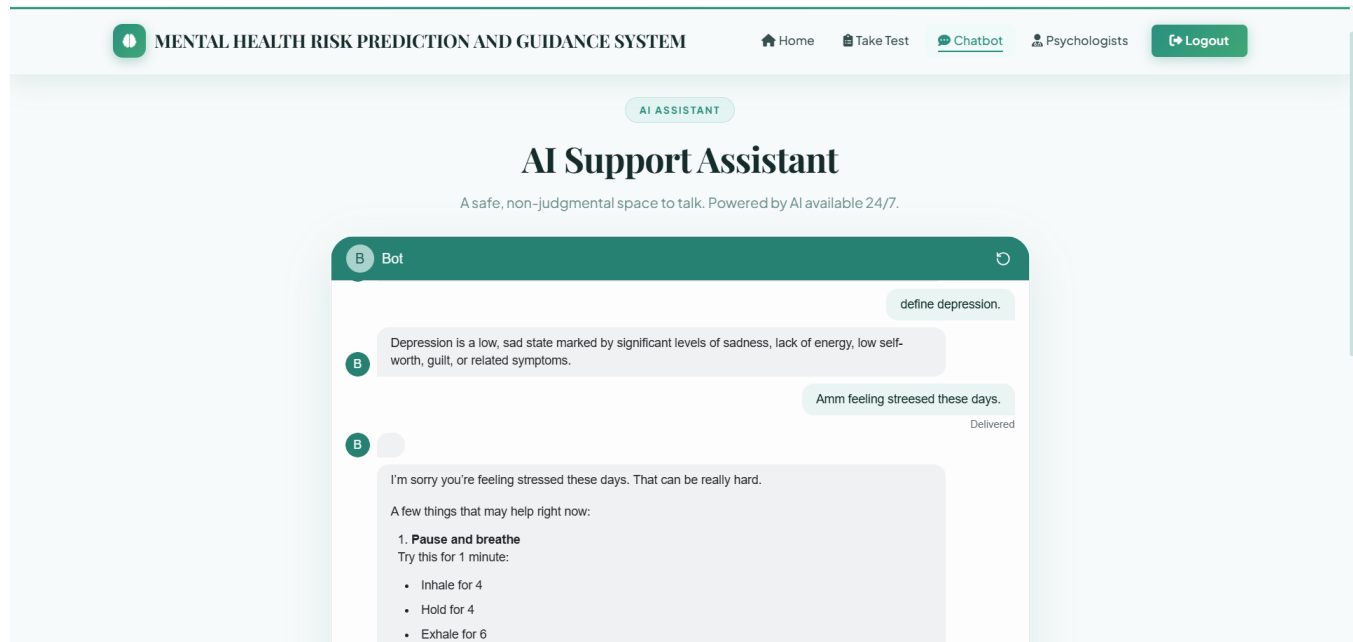
SUPPLEMENTARY LIFESTYLE-BASED RISK

Lifestyle Risk: High

This is a supplementary machine-learning indicator based on lifestyle factors. It is not a diagnosis.

8.1.5 Supportive Chatbot

The Botpress-based chatbot provides users with a supportive, non-judgmental space to ask questions about mental health. Grounded in a knowledge base from Abnormal Psychology (Comer & Comer), it provides psychoeducational responses, coping strategy information, and guidance about seeking professional help without diagnosing any condition.

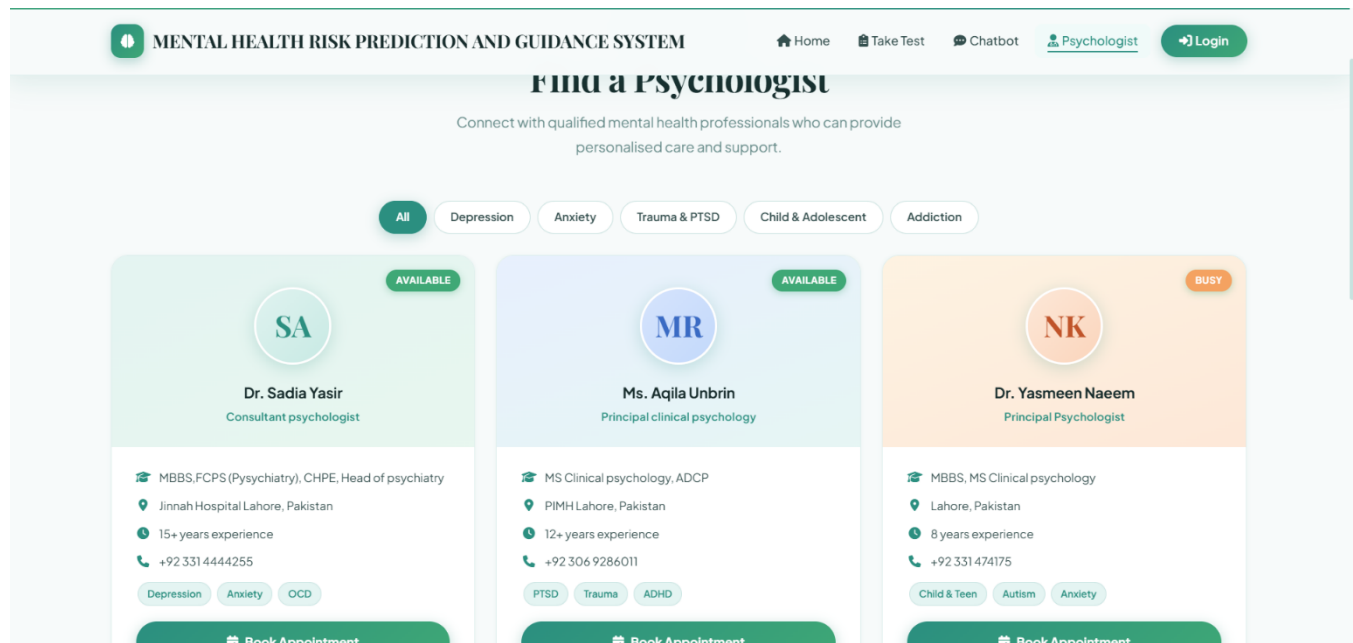


8.1.6 Responsive Frontend

The system is accessible through a web browser on desktop computers, tablets, and mobile phones. Bootstrap 5 ensures the layout adapts correctly to different screen sizes. The interface is clean, intuitive, and requires no technical knowledge to use.

8.1.7 Professional Guidance and Booking Pages

The system includes frontend pages displaying information about mental health professionals and appointment booking options, directing users who receive elevated assessment results toward professional help.



8.1.8 Crisis Helpline Page

The Crisis Helpline page serves as NEXORA's immediate emergency intervention hub designed for users experiencing acute distress or mental health crises. Built with high-contrast, clear layout elements to ensure zero friction during high-stress interactions, the interface prioritizes immediate speed of access over complex navigation. It features direct click-to-dial emergency numbers, crisis text line links, and quick-access safety planning guides alongside prominent safety disclaimers and persistent call-to-action buttons for urgent professional assistance.

8.1.9 Self-Help Articles Page

The Self-Help Articles page acts as the platform's primary psychoeducation repository, empowering users with evidence-based coping mechanisms and mental health knowledge. Content is organized into responsive card grids with clean typography and preview excerpts, allowing for efficient browsing across topics such as sleep hygiene, anxiety management, stress reduction, and academic burnout. The page incorporates a searchable library structure equipped with reading time indicators, category filtering, and direct links to self-assessment tools.

8.1.10 Terms of Use Page

The Terms of Use page establishes the legal framework, rules of engagement, and service boundaries governing the use of the NEXORA web application. Formatted with collapsible accordions and clean,

sectioned text for easy readability, the page clearly outlines platform usage policies, user account responsibilities, and intellectual property rights. Additionally, it features explicit medical disclaimers clarifying that NEXORA's AI tools offer risk guidance and educational insights rather than formal medical diagnoses.

8.1.11 Privacy Policy Page

The Privacy Policy page details NEXORA's data governance standards, user privacy protections, and compliance protocols regarding personal and psychometric information. Structured with a clean side-navigation index to enhance scannability and legal transparency, the document explicitly defines what data is gathered including assessment inputs, PHQ-9 scores, and usage logs. It thoroughly explains end-to-end data encryption, storage security, user rights regarding data deletion, and strict policies against third-party data sharing.

Chapter No. 9
**CONCLUSION AND FUTURE
WORK**

9.1 - Summary:

NEXORA the AI-Based Mental Health Risk Prediction and Guidance System was developed in response to a clearly identified and socially significant need, the absence of an accessible, responsible, and locally informed digital platform that could help a Pakistani individual experiencing symptoms of depression take an initial structured step toward understanding their situation and finding support.

The system addresses this need through a combination of components, each playing a distinct role. The PHQ-9 questionnaire, scored through a rule-based engine, provides the structured and clinically established foundation for the assessment. The Random Forest lifestyle-risk classifier adds supplementary context about the user's daily life circumstances. The Botpress-powered chatbot grounded in established psychological literature offers supportive, educational conversation. The Flask backend connects these components through a clean API. MySQL provides secure user account management. And the HTML/CSS/JavaScript/Bootstrap frontend presents everything in a clean, responsive, accessible interface.

Critically, the project was not developed solely as a technical exercise. The team visited Fatima Memorial Hospital, the Punjab Institute of Mental Health, and Jinnah Hospital, and the guidance received from healthcare professionals shaped the responsible design of the system most importantly, the decision to ground the assessment in the PHQ-9 rather than relying solely on AI, a recommendation that came directly from the medical professionals at Jinnah Hospital.

Throughout, NEXORA is positioned as a screening and guidance aid, not a diagnostic tool. Every component from the PHQ-9 engine to the chatbot to the results page is designed with this principle at its core.

9.2 - Requirements Traceability:

FR ID	Requirement	Implemented Feature
FR01/02	Registration & Authentication	MySQL nexora_db users table with hashed passwords; Flask /register and /login endpoints.
FR03/04	PHQ-9 Questionnaire & Scoring	Rule-based engine in Flask /predict endpoint; five severity categories per standard PHQ-9 bands.
FR05	Self-Harm / Suicide-Risk Flag	Rule engine checks PHQ-9 item 9; flag included in JSON response and displayed with crisis guidance.
FR06/07	Lifestyle Questions & Classification	Frontend encodes options 0–3; Flask passes to phq_risk_model.pkl; Random Forest returns Low/Moderate/High.
FR08	Results Display	Combined, clearly distinguished PHQ-9 and lifestyle-risk results with guidance text and disclaimer.
FR09	Chatbot	Botpress chatbot with Comer & Comer knowledge base and fallback for out-of-scope queries.
FR10/11	Professional Guidance / Booking	Frontend pages for psychologist information and appointment booking.

FR12	Logout	[Confirm session-termination implementation and mechanism]
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9.3 - Limitations:

- NEXORA is a screening and guidance tool, not a clinical diagnostic instrument. PHQ-9 results, lifestyle-risk classifications, and chatbot responses must never be interpreted as a medical diagnosis.
- PHQ-9 results are influenced by the honesty and accuracy of the user's self-reporting. Self-report measures are inherently subject to response bias.
- The Random Forest lifestyle-risk model's training data, methodology, and performance metrics have not been fully documented. Its outputs should be treated as supplementary context only.
- The Google Colab and ngrok development arrangement is not production-grade. Sessions time out and URLs change.
- The system is currently available in English only, limiting accessibility for Urdu-speaking users.
- Assessment history, doctor profiles, and appointment records are not currently persisted in the database in the confirmed current implementation.
- A native mobile application is not available.

9.4 - Future Roadmap:

The following enhancements are proposed for future versions of NEXORA. These are future possibilities and are not implemented in the current system.

9.4.1 Stable Production Deployment

The most immediate priority is migrating the Flask backend from Google Colab and ngrok to a stable, persistent hosting environment with a managed MySQL instance. This would make the system continuously available for real-world use.

9.4.2 Urdu-Language Support

Adding a fully translated Urdu-language interface is a high-priority enhancement, given that a significant proportion of the target Pakistani user population is more comfortable in Urdu.

9.4.3 Additional Validated Assessment Instruments

The assessment module could be extended to include the GAD-7 for anxiety, the PCL-5 for PTSD, or instruments for bipolar disorder transforming NEXORA into a broader mental health screening platform.

9.4.4 Improved Machine-Learning Models

The Random Forest lifestyle-risk model could be improved through larger and better-quality training datasets, more features, and exploration of other algorithms. Full documentation of training methodology and performance metrics is also required.

9.4.5 Chatbot Behavioural and Facial Analysis

Future chatbot versions could incorporate analysis of the user's communication patterns sentiment, pace, and content of messages as additional emotional state indicators. More advanced versions could potentially include facial expression analysis through camera input, with appropriate ethical safeguards and explicit user consent.

9.4.6 Full Appointment Booking Backend

A future version should implement a complete backend for the professional guidance and booking features, including database tables for doctor profiles and appointment slots, real-time availability management, and booking confirmation notifications.

9.4.7 Doctor and Patient Dashboards

A dedicated dashboard for doctors to view patient assessment histories, manage appointment slots, and track progress. Users could have a personal history dashboard showing assessment results over multiple sessions.

9.4.8 Mobile Application

A native mobile application for Android and iOS would significantly increase the system's accessibility, particularly in Pakistan where smartphone usage often exceeds desktop usage.

9.4.9 Multilingual Support and Wearable Integration

Beyond Urdu, future versions could support other regional languages. Integration with wearable devices could provide physiological data (sleep quality, heart rate variability, activity levels) as additional lifestyle-risk features, enabling more passive and continuous monitoring.

9.4.10 Live Video Session & Behavioral Analysis Integration

In future iterations of the NEXORA platform, the system will incorporate live video consultation capabilities to facilitate real-time virtual therapy sessions between users and verified psychologists. This feature will allow mental health professionals to directly observe non-verbal cues, facial expressions, and subtle behavioral indicators during evaluation sessions. Furthermore, this live video layer can be enhanced with computer-vision-assisted affective computing to assist practitioners by detecting micro-expressions and tracking emotional fluctuations over time, thereby providing deeper clinical insights to support comprehensive mental health assessments.

9.4.11 Expansion to Other Disease Domains

During the meeting with the Head of Department at Jinnah Hospital, an important future possibility was discussed: the potential for expanding the system architecture beyond mental health and into other disease domains. The Head of Department expressed interest in a system that could support structured assessment and guidance for other medical conditions using the core NEXORA architecture as a template adaptable for different clinical specialties.

This possibility validates the scalability of the NEXORA design. The fundamental pattern a validated structured questionnaire scored by a rule engine, supplemented by a machine-learning model, with a

supportive guidance layer and professional referral is applicable in principle to many clinical contexts, such as cardiovascular risk screening, diabetes risk assessment, or condition-specific patient education.

These are future possibilities only and are not implemented in the current version of NEXORA. However, the expressed professional interest from Jinnah Hospital demonstrates real-world confidence in the potential of the system's architecture.

9.5 - Use Cases:

Use Case 1: User Registration

Use Case Name	User Registration / Create Account
Actor(s)	New User / Patient
Preconditions	User has not previously registered, registration page is accessible.
Main Flow	1. User opens the registration page. 2. User enters full name, email address, and password. 3. Frontend submits data to Flask backend. 4. Backend validates input. 5. Backend checks whether email already exists. 6. Backend hashes the password. 7. Backend inserts new user record into nexora_db users table. 8. Frontend displays success and redirects to login page.
Postconditions	New user record in users table. User can now log in.
Alternative Flows	If email already registered: backend returns error; frontend displays appropriate message. If required fields missing: frontend validation prevents submission.

Use Case 2: User Login & Authentication

Use Case Name	User Login & Authentication
Actor(s)	Registered User / Admin
Preconditions	Account exists in the system database.
Main Flow	1. Actor opens login.html and inputs registered credentials (email & password). 2. System compares hashed input credentials against MySQL database records. 3. System validates account credentials and generates an authenticated user session. 4. System redirects actor to their respective dashboard (User Assessment, Psychologist Schedule, or Admin Panel).
Postconditions	Active user session token created; actor is granted role-based access.
Alternative Flows	Invalid Credentials: System shows error "Invalid email or password" without revealing existing usernames.

Use Case 3: Submit Mental Health & Lifestyle Risk Assessment

Use Case Name	User Registration / Create Account
Actor(s)	Registered User / Patient
Preconditions	The assessment page is accessible
Main Flow	1. User opens assessment page. 2. User answers 9 PHQ-9 items and 3 lifestyle questions. 3. Frontend validates all questions are answered. 4. Frontend encodes lifestyle selections (0–3). 5. Frontend submits data to POST /predict. 6. Backend validates input; computes PHQ-9

	result via rule engine. 7. Backend obtains lifestyle-risk classification from Random Forest model. 8. Backend returns combined JSON result. 9. Frontend displays results, clearly distinguishing PHQ-9 from lifestyle-risk. 10. Appropriate guidance and disclaimer displayed.
Postconditions	Results displayed to user
Alternative Flows	If not all questions answered form validation prevents submission. If backend unavailable frontend displays an error message.

Use Case 4: Interact with AI Guidance Chatbot

Use Case Name	Interact with AI Guidance Chatbot
Actor(s)	General User
Preconditions	Web application initialized
Main Flow	1. User opens the chat widget embedded in the platform interface, 2. User submits text queries or selects pre-defined intent buttons (e.g., "Manage Stress", "Take Assessment"). 3. System routes the input through Botpress workflow rules and AI natural language processing. 4. Chatbot generates contextual responses, psychoeducational tips, or redirects the user to relevant site sections (Crisis.html, article.html)
Postconditions	Conversational guidance, coping strategies, or feature redirects are provided in real-time.
Alternative Flows	High-Risk Keyword Trigger: If crisis keywords (e.g., self-harm) are detected, the chatbot overrides standard dialogue and immediately prompts crisis intervention links

Use Case 5: Access Crisis Helpline & Emergency Resources

Use Case Name	Access Crisis Helpline & Emergency Resources
Actor(s)	General User
Preconditions	User opens crisis.html directly or via automated redirect.
Main Flow	User enters the emergency intervention hub, User selects a helpline number according to the quandary for immediate phone or messaging connection.
Postconditions	Emergency support numbers or text lines are placed within one-click access.

Use Case 6: Browse & Book Psychologist Consultation

Use Case Name	Browse & Book Psychologist Consultation
Actor(s)	General User / Patient
Preconditions	User navigates to psychologists.html.
Main Flow	1. User browses or filters verified mental health professionals by specialization, availability, or language. 2. User selects a practitioner profile and chooses an available session slot. 3. User provides basic contact details and confirms appointment request. 4. System logs the booking into the MySQL database and dispatches notification alerts to the practitioner.
Postconditions	A consultation booking request is logged and communicated to the selected professional.

Use Case 7: Platform Governance & Data Administration

Use Case Name	Platform Governance & Data Administration
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Actor(s)	System Administrator
Preconditions	Administrator possesses authorized, direct access credentials to the underlying database management layer (e.g., MySQL).
Main Flow	Administrator authenticates securely into the MySQL database engine. Admin updates or inserts educational resource records for articles.html and emergency contact records for crisis.html directly within database tables. Admin reviews user assessment logs, risk predictions, and system security logs. Database executes the update queries and commits changes across the NEXORA web application.
Postconditions	Practitioner verification statuses are updated, or platform content and system logs are directly updated in the database tables.